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Key Points:

- A deep learning method with adaptively weighted loss is proposed for 1-D homogeneous soil water flows
- Adaptively weighted loss method helps physics-informed neural network focus more on greater loss regions to improve the accuracy
- The soil water flows and constitutive relationships of various soils can be accurately predicted via the proposed method

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A Physics-Informed Deep Learning Method With Adaptively Weighted Loss for Modeling Soil Water Flows

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Abstract Richards' equation, widely used to model soil water flows, presents numerical challenges due to the high nonlinearity of its constitutive relationships. The deep learning method with a physics-informed neural network (PINN) provides a fresh perspective for solving this equation without prior knowledge of soil water constitutive relationship. However, existing PINN-based methods for Richards' equation are still significantly limited by the feasible soil types, and further developments are urgently needed to make the neural network models practically applicable to various types of soils. In this paper, we introduce a deep learning method, "PINN-AWL," which simultaneously build the PINN model for prediction of soil water flows and establish the constitutive relationships between soil matric potential, soil water content and unsaturated hydraulic conductivity. An adaptively weighted loss is specially designed for the training process of this model. Specifically, the loss function is adjusted with self-adaptive weights at the training points in each iteration of training. This allows the proposed PINN-AWL to automatically focus more on regions where the solution is difficult to fit. The prediction accuracy and generalization ability of the PINN-AWL are thoroughly tested on various soils ranging from silty to sandy types. We also conduct studies to investigate the optimal structure and hyper-parameters used in the proposed method. The numerical results demonstrate that the proposed PINN-AWL significantly outperforms both the standard PINN and the monotonic PINN, especially on soils exhibiting strong nonlinearity in constitutive relationships, as indicated by larger "n" values in the van Genuchten-Mualem model.

1. Introduction

Accurate soil water prediction is important for hydrological cycling simulation, agriculture water management, drought and waterlogging disaster regulation (Chao et al., 2022; L. Liu et al., 2020; Seneviratne et al., 2010). Richards' equation has been popularly used to model the soil water dynamics, which consists of two key variables, for example, the soil matric potential ψ and the volumetric soil water content θ (Richards, 1931). There exist many numerical methods developed for Richards' equation, such as finite difference method (FDM), finite volume method (FVM) and finite element method (FEM) (Chávez-Negrete et al., 2018; Eymard et al., 1999; Pour et al., 2011; Xie et al., 1998). The major difficulty for numerically solving Richards' equation is that the exact priori knowledge of the constitutive relationships between the soil matric potential and the soil water content and between the soil matric potential and the unsaturated hydraulic conductivity are required (Farthing & Ogden, 2017). Some empirical models have been proposed to characterize the constitutive relationships of the soil water equation (Brooks & Corey, 1964; Liao et al., 2018; Mualem, 1976; Tuller & Or, 2001; van Genuchten, 1980). Among them the van Genuchten-Mualem (VGM) model is the most popular one, which is considered to be quite accurate in describing the two nonlinear constitutive relationships as shown in Figure 1. There are six parameters included in van Genuchten-Mualem model, and sensitivity analysis shows that at least four of which are sensitive ones, which are θ_r , α , n , θ_s (Chen et al., 2016). Specifically, θ_r is the residual volumetric water content, α is an empirical parameter related to the inverse of the air-entry suction, n is a pore-size distribution index, θ_s is the saturated volumetric soil water content. Intensive labor work must be done to obtain the soil water characteristics curves (Kanzari et al., 2020; Londra, 2010). Also, the well-established laboratory methods are not applicable to field scales due to scale mismatches (Bandai & Ghezzehei, 2021; Romano et al., 2002). Thus, it is still practically a great challenge to solve the Richards' equation and obtain the constitutive relationships simultaneously.

The physics-informed neural network (PINN) integrates prior physical laws into the loss function for training the neural network, ensuring that the trained network satisfies the governing equations. PINN has been utilized to

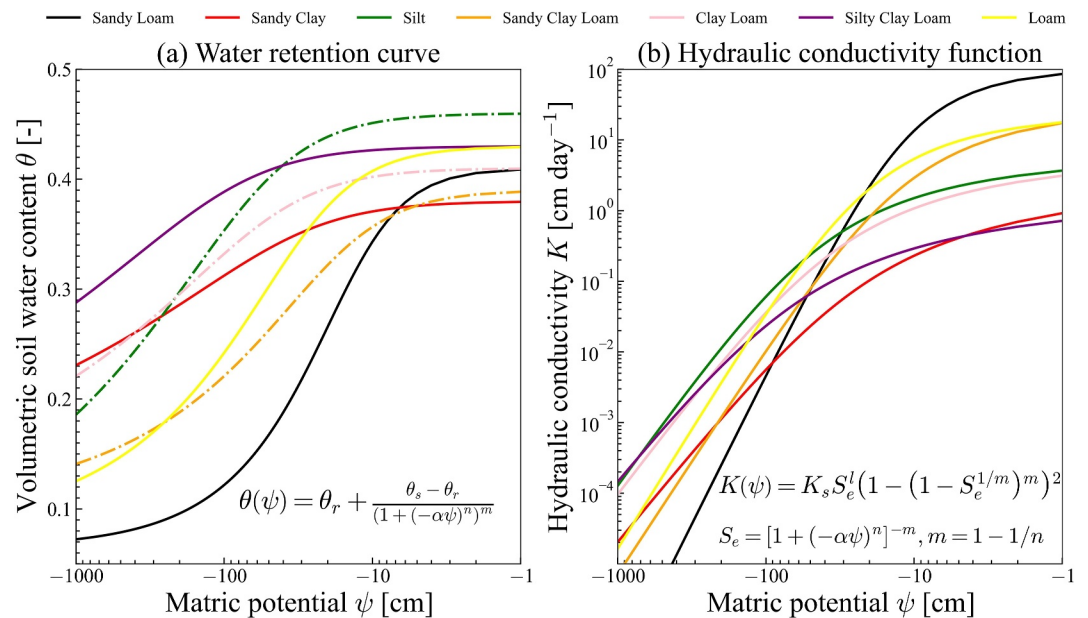


Figure 1. Constitutive relationships ((a) water retention curve and (b) hydraulic conductivity function) for different soil types generated by the van Genuchten-Mualem model (Mualem, 1976; van Genuchten, 1980). It should be noted that the parameters used are from Carsel and Parrish (1988). θ_s is the saturated soil water content; θ_r is the residual soil water content; K_s is the saturated hydraulic conductivity; α , n , and l are the empirical parameters in the soil water retention function.

solve various types of partial differential equations (PDEs), including heat transfer equation (Zobeiry & Humfeld, 2021), biharmonic equations of elasticity (Vahab et al., 2022), and Navier-Stokes equations (Jin et al., 2021). Moreover, PINN has demonstrated its robust capability in solving inverse problems, such as supersonic flows (Jagtap et al., 2022), natural-state geothermal systems (Ishitsuka & Lin, 2023), fluid mechanics problems (S. Xu et al., 2023) and multiphase thermo-hydro-mechanical processes in porous media (Amini et al., 2023).

In hydrology applications of PINN can be found in both forward and inverse problems, such as predicting streamflow (Bojović et al., 2022; Zhang et al., 2022) and large-scale river modeling (Feng et al., 2023). The groundwater equation was successfully solved by PINNs in N. Wang et al. (2020) and R. Xu et al. (2021). However, directly applying PINN to Richards' equation has not yielded satisfactory results due to the highly nonlinear nature of its constitutive relationships. Tartakovsky et al. (2020) adopted PINN to estimate the hydraulic conductivity of saturated and unsaturated soils using the two-dimensional time-independent Richardson-Richards equation (RRE) and the capillary pressure measurements. Bandai and Ghezzehei (2021) proposed a deep learning framework consisting of three fully-connected neural networks for rough estimation of water retention curve and hydraulic conductivity function based on the one-dimensional Richards' equation. To obtain accurate results, monotonicity was imposed by constraining the weights of the latter two networks to be non-negative. However, soil water characteristic curves are not simply monotonic as shown in Figure 1, and the curve shape varies among different soils. Introducing non-negative weights to the neural network is insufficient to describe the features of soil moisture characteristic curves. As a result, the constitutive relationships derived from the study by Bandai and Ghezzehei (2021) fail to produce satisfactory outcomes in the wet end. The research conducted by Depina et al. (2022) demonstrated that PINNs can solve the inverse problem, providing a close approximation to the solutions of the Richards' equation and estimating the parameters of the van Genuchten-Mualem model, but fewer scenarios and soil types were studied. A multi-physics information neural network (Y. Wang et al., 2023) was further proposed to improve soil water results by adding soil thermal information. Despite its promising results, this approach required additional soil heat data was primarily tested on a limited range of soil types, leaving space for further improvement. Elkhadrawi et al. (2024) developed a novel PINN approach to estimate hydraulic conductivity based on volumetric soil water content measurements from three specific locations, positioning their work in direct comparison with the studies conducted by Bandai and Ghezzehei (2021), this research was limited to only three types of soil, and the predictive performance for some soils was not entirely satisfactory. Therefore, designing a robust PINN that can capture various soil water

variations and accurately estimate constitutive relationships from limited volumetric soil water content data still remains a challenging task.

Capturing exact solutions in regions with sharp changes could be key to improving training accuracy. An efficient neural network could focus on regions with large errors and attempt to minimize the loss in these regions using various methods, such as sampling strategies and loss functions. For instance, Zhang et al. (2022) adapted a local refined sampling strategy near pumping wells to capture drastic hydraulic head changes. The residual-based adaptive refinement method was proposed for PDEs that exhibit solutions with steep gradients (Lu et al., 2019). An adaptive sampling strategy by adjusting the weights of training points was proposed for high-dimensional partial differential equations to improve accuracy (Gu et al., 2021). The gradient-enhanced method was proposed to embed the information of PDE residual into the loss function to improve prediction results (Yu et al., 2022). Introducing a self-adaptive procedure could be a potential approach for establishing a robust PINN for solving soil water flow equations.

In this paper, we propose a novel deep learning method, “PINN-AWL,” for the solution of Richards' equation which utilizes the PINN architecture and a specially designed adaptively weighted loss. In each training iteration, self-adaptive weights are calculated based on the loss value of each training point. Regions with large errors, such as mutation interfaces and other challenging areas, are automatically given larger weights in the loss function, prompting the network to improve on those points. Minimizing losses and maximizing weights simultaneously helps train the network more efficiently. The fundamental principle of the proposed model allows PINN to pay more attention to points with large losses, making it easier to learn soil water profiles that usually exhibit sharp changes. Hydrus-1D (Simunek et al., 2013) is used to simulate the movement of water in different soils to generate data for training and model evaluation. The accuracy and generalization capability of our PINN-AWL for solving Richards' equation are demonstrated with examples of various soil types and different upper boundary conditions and comparisons with the standard PINN and the monotonic PINN. We also investigated the effects of hyper-parameters used for estimating the adaptively weighted loss, and the influence of spatial sparsity and noise of the training data on the performance of our PINN-AWL.

The remainder of this paper is organized as follows. In Section 2, we briefly review Richards' equation and deep neural networks. Section 3 offers a detailed description of our proposed method, PINN-AWL, including the network architecture, the loss function, and the training process. Evaluation measurements and testing examples are illustrated in Section 4, and Section 5 presents the test results to demonstrate the outstanding performance of our PINN-AWL under different circumstances. Section 6 further discusses the effect of the adaptively weighted loss. In Section 7, we address the performance of PINN-AWL on sparse and noisy training data. Finally, some concluding remarks are drawn in Section 8.

2. Preliminary

2.1. Richards' Equation

Richards' equation of one-dimensional uniform soil water flow, ignoring the lag term and the source-sink term, is given by,

$$\frac{\partial \theta}{\partial t} = \frac{\partial}{\partial z} \left[K \left(\frac{\partial \psi}{\partial z} + 1 \right) \right], \quad (1)$$

where θ [$\text{L}^3 \text{L}^{-3}$] is the volumetric soil water content; t [T] is the time; z [L] is the vertical coordinate; K [L T^{-1}] is the hydraulic conductivity, and ψ [L] is the matric potential of soil water. Our study focuses on developing a robust Physics-Informed Neural Network (PINN) capable of accurately capturing various soil water variations and estimating constitutive relationships from Richards' equation using limited volumetric soil water content data. Consequently, we do not incorporate initial and boundary conditions into Equation 1. As mentioned above, the volumetric soil water content and hydraulic conductivity can be expressed as functions of the matric potential, respectively. The van Genuchten-Mualem model (Mualem, 1976; van Genuchten, 1980) is applied to represent

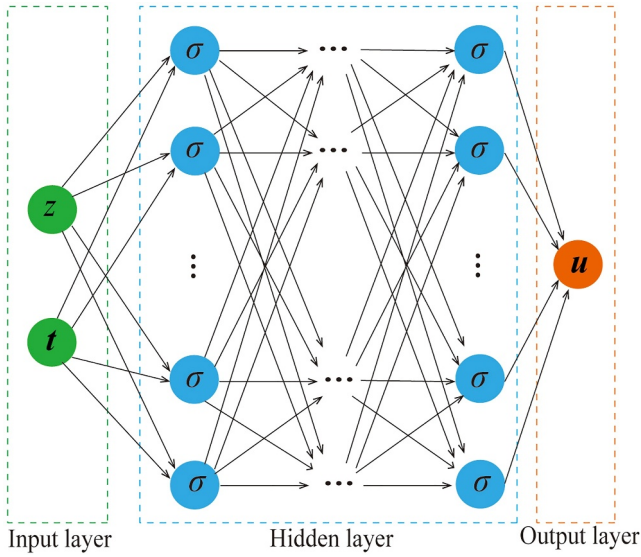


Figure 2. An example of the fully-connected neural network, where $x = (z, t)$ is the input; σ is the activate functions, and u is the output.

2.2. Deep Neural Networks

Deep neural networks have been popularly used to learn the functional relationship between input samples and output targets. In this paper, we consider fully-connected neural network (FCNN) (as shown in Figure 2), which is also the backbone of PINN. Herein, we will briefly introduce how the network works. Let $f^L(x)$ be the L layers of neural network of $\mathbb{R}^{\text{in}} \rightarrow \mathbb{R}^{\text{out}}$ and n_l denote the number of neurons in the l -th layer for $l = 1, 2, \dots, L$. Thus, each layer has the weight matrix $\mathbf{W}^l \in \mathbb{R}^{n_l \times n_{l-1}}$ and the bias vector $\mathbf{b}^l \in \mathbb{R}^{n_l}$. The output of the previous layer is the input of the current layer, and the fully-connected feedforward neural network can be defined as follows,

$$\begin{cases} f^0(\mathbf{x}) = \mathbf{x} = (x, t) \\ f^l(\mathbf{x}) = \sigma(\mathbf{W}^l f^{l-1}(\mathbf{x}) + \mathbf{b}^l), \text{ for } 1 \leq l \leq L-1, \\ f^L(\mathbf{x}) = \mathbf{W}^L f^{L-1}(\mathbf{x}) + \mathbf{b}^L \end{cases} \quad (5)$$

where $\mathbf{x} = (x, t)$ are the inputs; $f^0(\mathbf{x})$ is the output of input layer; $f^{l-1}(\mathbf{x})$ and $f^l(\mathbf{x})$ are the input and output of the l -th layer, respectively; $f^L(\mathbf{x})$ is the output of neural network, and σ is the activation function. Activation functions are key components of the neural network, which determine the nonlinear representation of the network. Some typical activation functions, for example, the rectified linear activation (ReLU), logistic sigmoid, hyperbolic tangent (Tanh), are commonly used. The Tanh function was used in this study. The set of weight matrixes $\mathbf{W}^l$ and bias vectors $\mathbf{b}^l$ of each layer, $\{\mathbf{W}^l, \mathbf{b}^l\} l = 1, L$, are trainable parameters of the neural network, which can be obtained by minimizing the loss function.

3. A Physics-Informed Deep Learning Method for Modeling Soil Water Flows

3.1. Network Architecture

We propose a deep learning method, “PINN-AWL,” which build the PINN model for prediction of soil water flows and establish the constitutive relationships for Richards' equation simultaneously. In particular, an adaptively weighted loss is specially designed for the training process of the PINN-AWL. The architecture of our PINN-AWL is shown in Part (1) of Figure 3, which contains a subnetwork for predicting the soil matric potential $\hat{\psi}$, followed by two subnetworks to predict the volumetric soil water content $\hat{\theta}$ and the unsaturated hydraulic conductivity $\hat{K}$ from the collocation points (t, z) and the time-dependent volumetric soil water content data.

the functions $\theta(\psi)$ and $K(\psi)$ when using Hydrus-1D, we then have the following expressions for the constitutive relations,

$$\theta(\psi) = \begin{cases} \theta_s, & \psi \geq 0 \\ \theta_r + \frac{\theta_s - \theta_r}{(1 + (-\alpha\psi)^n)^m}, & \psi < 0 \end{cases} \quad (2)$$

and

$$K(\theta(\psi)) = \begin{cases} K_s, & \psi \geq 0 \\ K_s S_e^l \left(1 - (1 - S_e^{1/m})^m\right)^2, & \psi < 0 \end{cases} \quad (3)$$

where θ_r [$\text{L}^3 \text{L}^{-3}$] is the residual water content; θ_s [$\text{L}^3 \text{L}^{-3}$] is the saturated water content; K_s [L T^{-1}] is the saturated hydraulic conductivity; α [L^{-1}], n [–], m [–] and l [–] are empirical shape parameters, and m is related to n as $m = 1 - 1/n$; S_e [–] is the effective saturation defined as,

$$S_e = \frac{\theta - \theta_r}{\theta_s - \theta_r}. \quad (4)$$

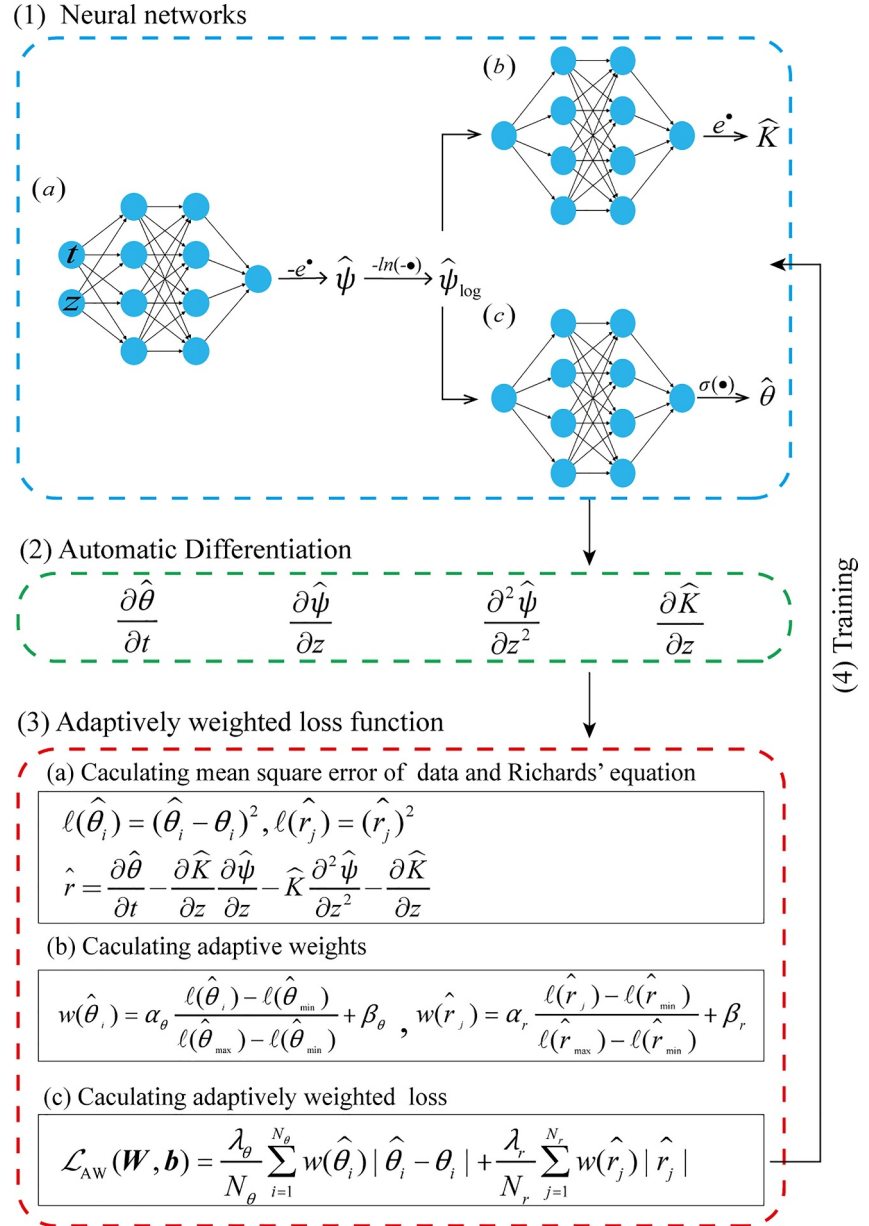


Figure 3. Network architecture and the pipeline of the PINN-AWL for prediction of soil water flows with simultaneous learning of the constitutive relationships of Richards' equation. Note that (t, z) is the input; $\hat{\psi}, \hat{\theta}, \hat{K}$ are the soil matric potential, the volumetric soil water content, the unsaturated hydraulic conductivity predicted by the network, respectively; the symbol “ $\cdot$ ” in the function (e.g., $-e^*$, $-\ln(\cdot)$, e^* , and $\sigma(\cdot)$) denotes the input, and $-e^*$, $-\ln(\cdot)$, e^* , and $\sigma(\cdot)$ represent the transformations of the input using negative exponential function, negative logarithmic function, exponential function and sigmoid function, respectively; $\alpha_r, \beta_r, \alpha_{\theta}, \beta_{\theta}$ are the hyper-parameters used in calculating adaptive weights.

We first introduce the neural network for learning the soil matric potential,

$$\hat{\psi}(t, z) = \hat{f}_{\psi}(t, z; \mathbf{W}_{\psi}, \mathbf{b}_{\psi}), \quad (6)$$

where $\mathbf{W}_{\psi}$ and $\mathbf{b}_{\psi}$ are the sets of weight and bias parameters in this network. To guarantee the matric potential $\hat{\psi}(t, z)$ to be negative, the negative exponential function is added to the output layer of $\hat{f}_{\psi}$. Ensuring the matric

potential remains negative is specific to unsaturated flow and is not sufficient to describe saturated flow. Thus, the PINN-AWL should only be applied to data sets that the soil remains unsaturated, and this can be achieved by observing distribution of the volumetric soil water content data and selecting appropriate data set. Next are two subsequent subnetworks used to model the two constitutive relationships, whose inputs are the logarithmic transformation of the output $\hat{\psi}(t, z)$ from Equation 6, that is,

$$\hat{K}(\hat{\psi}) = \hat{f}(\hat{\psi}_{\log}(t, z; \mathbf{W}_{\psi}, \mathbf{b}_{\psi}); \mathbf{W}_K, \mathbf{b}_K), \quad (7)$$

$$\hat{\theta}(\hat{\psi}) = \hat{f}(\hat{\psi}_{\log}(t, z; \mathbf{W}_{\psi}, \mathbf{b}_{\psi}); \mathbf{W}_{\theta}, \mathbf{b}_{\theta}), \quad (8)$$

where the use of $\hat{\psi}_{\log} = -\ln(-\hat{\psi})$ can help train the entire network; $\mathbf{W}_K$ and $\mathbf{b}_K$ are the sets of weight and bias parameters in the subnetwork for learning the unsaturated hydraulic conductivity; $\mathbf{W}_{\theta}$ and $\mathbf{b}_{\theta}$ are the sets of weight and bias parameters in the networks for learning volumetric soil water content. Exponential function and sigmoid function are respectively used in the output layers of $\hat{f}_K$ and $\hat{f}_{\theta}$ to ensure that the unsaturated hydraulic conductivity is non-negative and the value of volumetric soil water content falls between 0 and 1.

3.2. Loss Functions

A key component determining the effectiveness of the proposed PINN-AWL (containing three subnetworks (Equations 6–8)) is the selection of appropriate loss function. Since our task is to solve the Richards' equation and learn the constitution relationships simultaneously, the loss function will consist of two parts: one is the residual error of the Richards' equation and the other is the error between the predicted and true volumetric soil water content data. We will set up and compare two choices for the loss function as shown below.

It is worth highlighting that our loss function omits the boundary and initial conditions and only contains the Richards' equation and the volumetric soil water content observations. Excluding initial and boundary conditions in the loss function has several advantages. Firstly, the absence of boundary and initial conditions simplifies the optimization problem and allows PINNs to focus on extracting soil moisture variation characteristics directly from volumetric moisture content data and Richards' equation. Secondly, it enhances the adaptability of the PINN, encouraging PINN to develop a more general understanding of the behavior of the system and learn the hydrodynamic properties of various soils. Thirdly, this feature enables the PINN to directly process a variety of data sets, which can correspond to different boundary conditions, without specifying boundary conditions. In summary, the deliberate exclusion of initial and boundary conditions from the loss function makes PINN adaptable and generalizable for modeling soil water flow.

3.2.1. Standard Loss Function

A natural choice is to use the mean square error (MSE) as the loss function, so the standard loss function including only the volumetric moisture content data term and the Richards' equation which can be expressed as,

$$\mathcal{L}_{\text{MSE}}(\mathbf{W}, \mathbf{b}) = \frac{\lambda_{\theta}}{N_{\theta}} \sum_{i=1}^{N_{\theta}} \ell(\hat{\theta}_i) + \frac{\lambda_r}{N_r} \sum_{j=1}^{N_r} \ell(\hat{r}_j), \quad (9)$$

with

$$\ell(\hat{\theta}_i) = (\hat{\theta}_i - \theta_i)^2, \ell(\hat{r}_j) = (\hat{r}_j)^2, \quad (10)$$

where $\mathbf{W} = (\mathbf{W}_{\psi}, \mathbf{W}_K, \mathbf{W}_{\theta})$ and $\mathbf{b} = (\mathbf{b}_{\psi}, \mathbf{b}_K, \mathbf{b}_{\theta})$ are the weights and biases of the PINN-AWL; N_{θ} is the number of the data samples $\{(t_i, z_i, \theta_i)\}$ with θ_i being the observed true volumetric soil water content at (t_i, z_i) ; N_r is the number of collocation points $\{(t_j, z_j)\}$ with $\hat{r}$ denoting the residual of the Richards' equation at (t_j, z_j) defined by,

$$\hat{r} = \frac{\partial \hat{\theta}}{\partial t} - \frac{\partial \hat{K}}{\partial z} \frac{\partial \hat{\psi}}{\partial z} - \hat{K} \frac{\partial^2 \hat{\psi}}{\partial z^2} - \frac{\partial \hat{K}}{\partial z}, \quad (11)$$

and $\lambda_\theta, \lambda_r$ are two hyper-parameters for balancing the two terms. The derivative terms of the above Equation 11 can be obtained by automatic differentiation (Part (2) of Figure 3). Unfortunately, due to the highly nonlinearity of Richards' equation, the adoption of Equation 9 may result in poor performance of the neural network in predicting soil water flow and learning the constitutive relationship of soil water flow.

3.2.2. Adaptively Weighted Loss

To better train the network and improve its performance, we propose to use an adaptively weighted loss, which aims to help the network focus on the points where the solution is difficult to fit during the training. For this purpose, we first introduce two pointwise weights, $w(\hat{\theta}_i)$ and $w(\hat{r}_j)$, where the former is applied to the volumetric soil water content of the i -th data sample and the latter is applied to the residual of the Richards' equation at the j -th collocation point. Specifically, they are defined as,

$$w(\hat{\theta}_i) = \alpha_\theta \frac{\ell(\hat{\theta}_i) - \ell(\hat{\theta}_{\min})}{\ell(\hat{\theta}_{\max}) - \ell(\hat{\theta}_{\min})} + \beta_\theta, \quad (12)$$

$$w(\hat{r}_j) = \alpha_r \frac{\ell(\hat{r}_j) - \ell(\hat{r}_{\min})}{\ell(\hat{r}_{\max}) - \ell(\hat{r}_{\min})} + \beta_r, \quad (13)$$

where $\ell(\hat{\theta}_i)$ and $\ell(\hat{r}_j)$ are the pointwise square errors defined in Equation 10; $\ell(\hat{\theta}_{\min})$ and $\ell(\hat{\theta}_{\max})$ are the minimum and maximum loss of volumetric soil water content in each epoch; $\ell(\hat{r}_{\min})$ and $\ell(\hat{r}_{\max})$ are the minimum and maximum loss of residuals of Richards' equation in each epoch of the training; $\alpha_r, \beta_r, \alpha_\theta, \beta_\theta$ are the empirical hyper-parameters for adjusting the weights. With the help of Equation 12 and Equation 13, the adaptively weighted loss function can then be constructed as follows (Part (3) of Figure 3),

$$\mathcal{L}_{AW}(\mathbf{W}, \mathbf{b}) = \frac{\lambda_\theta}{N_\theta} \sum_{i=1}^{N_\theta} w(\hat{\theta}_i) |\hat{\theta}_i - \theta_i| + \frac{\lambda_r}{N_r} \sum_{j=1}^{N_r} w(\hat{r}_j) |\hat{r}_j|. \quad (14)$$

To illustrate the effect of the adaptively weighted loss, we plot the surfaces of the MSE loss function (Equation 9) and the corresponding adaptively weighted loss function (Equation 14) in Figure 4 at the initial stage of the training process, from which we see that the adaptive weights can help amplify larger loss values while suppressing smaller loss values, thus making the training more focused on areas with larger loss values. Additional computational overhead is added to the calculation of adaptive weights, but the computational efficiency is not decreased greatly. The training time is basically the same as the standard PINN.

Similarly, Zong et al. (2023) introduced a normalized Physical Information Neural Network (PINN) weighting scheme. The normalization weighting method works by pre-determining a normalization factor based on the range or maximum value of the variables. It normalizes the variables in the PDE and assigns a weight to the loss term after normalizing loss term in the loss function. This carefully applied normalization and weighting technique improves the model's accuracy and robustness during learning and prediction without adding additional parameters. Although this framework was applied successfully in Zong et al. (2023), the nonlinearity of the Richards' equation makes it difficult to determine the range or maximum value of matric potential and hydraulic conductivity when only the volumetric water content is known. As a result, normalizing the matric potential, hydraulic conductivity, and the residual loss term becomes a significant challenge in this study.

Compare with the normalization PINN weighting scheme proposed by Zong et al. (2023), the PINN-AWL weighting scheme requires more parameters. The process of identifying suitable parameters is more complex

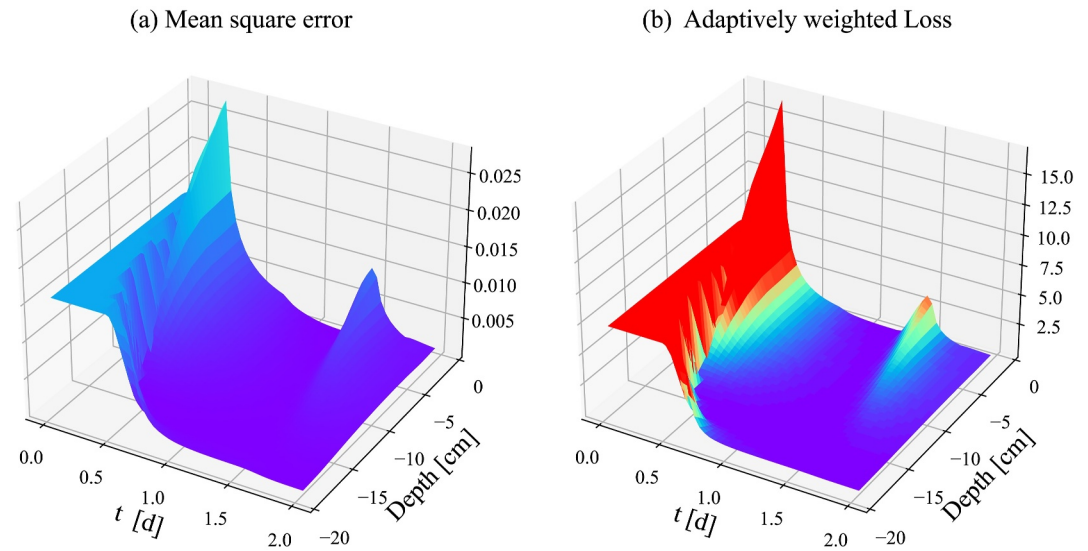


Figure 4. Plot of the surfaces of the MSE (Equation 9) and the adaptively weighted loss (Equation 14) at the initial stage of the training process: (a) mean square error, (b) adaptively weighted loss.

compared to the normalization scheme of Zong et al. (2023). However, PINN-AWL allows more flexibility in controlling the model's focus on regions with larger loss values by manually adjusting the parameters used to calculate the weights, which can improve accuracy and robustness. The weighting scheme of PINN-AWL is easy to understand and implement, which can be used jointly with other PINN model. In addition, the proposed PINN-AWL is compared with two adaptive networks: the Self-adaptive Physics-Informed Neural Network (SA-PINN) proposed by McClenny and Braga-Neto (2023), and the Learning Rate Annealing Physics-Informed Neural Network (LA-PINN) introduced by S. Wang et al. (2021). Detailed descriptions and corresponding results are provided in Appendix A.

Remark 1 PINN solves a PDE by embedding it in a loss function via automatic differentiation. MSE is the standard loss function used in PINN and thus the PINN with the MSE loss is referred to as the standard PINN. The standard PINN for Richards' equation adopts MSE to incorporate PDE residuals and data errors into the loss function. If there are some known features in the PDE, it is also possible to encode them into the network to enhance the network's ability to solve the PDE, such as monotonicity. Bandai and Ghezzehei (2021) proposed the monotonic PINN which restricts the weight parameters $\mathbf{W}_K$ and $\mathbf{W}_\theta$ to be non-negative to represent the monotonicity of the water retention curve (WRC) and the hydraulic conductivity function (HCF).

3.3. Training Process

In this section, we describe the training process of the proposed PINN-AWL. The purpose is to minimize the loss function $\mathcal{L}_{AW}(\mathbf{W}, \mathbf{b})$ (Equation 14) to find the best network parameters. Firstly, we need to set up the inputs to the network and the hyper-parameters of the network, such as the number of layers of the network, the number of neurons per layer and the learning rate. Secondly, the satisfactory parameters ($\mathbf{W}, \mathbf{b}$) will be obtained by minimizing the adaptively weighted loss function through a stochastic gradient descent method. In this paper, we choose Adam optimizer (Kingma & Ba, 2014) followed by LBFGS optimizer (D. C. Liu & Nocedal, 1989). The detailed training process is shown in Algorithm 1.

Algorithm 1. Training of PINN-AWL

Data: The architecture of PINN-AWL (the number of layers, the dimensionality of each layer, and the activation function); the number of training step $S=S_1+S_2$ with S_1 for Adam optimizer, S_2 for LBFGS optimizer; the collocation points $(t_j, z_j)_{j=1}^{N_c}$ sampled inside the domain; the data points $(t_i, z_i, \theta_i)_{i=1}^{N_d}$ at fixed depth; the learning rate lr ; the hyper-parameters λ_θ and λ_r for balancing loss terms; hyper-parameters $(\alpha_r, \beta_r, \alpha_\theta, \beta_\theta)$ used in calculating the adaptive weights.

Result: Find the best model with parameters (W, b) , which consist of (W_ψ, b_ψ) of sub-network $\hat{\psi}(t, z)$, (W_θ, b_θ) of sub-network $\hat{\theta}(\hat{\psi})$, and (W_K, b_K) of sub-network $\hat{K}(\hat{\psi})$.

```

1 Initialize network's parameters  $(W, b)$ ;
2 While  $s \leq S$  do
3   Calculate the data mean squared error  $\ell(\hat{\theta}_i)$  at each data point  $(t_i, z_i, \theta_i)$  and the PDE residual  $\ell(\hat{r}_j)$  at each collocation point  $\{(t_j, z_j)\}$  according to Equation 10;
4   Calculate the data weights  $\{w(\hat{\theta}_i)\}$  and the PDE residual weights  $\{w(\hat{r}_j)\}$  according to Equation 12 and Equation 13;
5   Calculate the sum of the total adaptively weighted loss  $\mathcal{L}_{AW}$  according to Equation 14;
6   if  $s \leq S_1$  then
7     update the parameters  $(W, b)$  via Adam optimizer
8   end if
9   if  $S_1 < s \leq S_2$  then
10    update the parameters  $(W, b)$  via LBFGS optimizer
11  end if
12 end while

```

4. Evaluation Measurements and Testing Examples

To evaluate the performance of PINN-AWL over the entire spatial and temporal scale, we compute the relative L_2 errors of volumetric soil water content, soil matric potential, hydraulic conductivity, and soil water flux density between the predictions by PINN models (including the standard PINN, monotonic PINN and the proposed PINN-AWL) and the reference solutions calculated by Hydrus-1D as follows,

$$\varepsilon^y = \sqrt{\frac{\sum_{t \in [0, T]} \sum_{z \in [-L, 0]} |\hat{y}(t, z) - y(t, z)|^2}{\sum_{t \in [0, T]} \sum_{z \in [-L, 0]} |y(t, z)|^2}}, \quad (15)$$

where $\hat{y}(t, z)$ is the predicted solution by the PINN models, and $y(t, z)$ is the reference solution. The temporal interval, denoted as $[0, T]$, is defined as the period ranging from 0 to 2 days, with a temporal resolution of 0.01 days. This resolution allows us to capture the dynamics of the system with sufficient detail over the specified time frame. The spatial interval, represented as $[-L, 0]$, denotes the range from -20 to 0 cm, with a spatial discretization of 0.1 cm. This discretization provides a balance between the level of detail in the spatial distribution and computational efficiency. The predicted soil water flux density can be obtained by Buckingham-Darcy law (Buckingham, 1907),

$$\hat{q} = -\hat{K} \left(\frac{\partial \hat{\psi}}{\partial z} + 1 \right). \quad (16)$$

The vertical soil water flow in a 100 cm soil column within 2 days under eight different soil types (sandy loam, sandy clay, silt, silt loam, sandy clay loam, clay loam, silty clay loam, loam) are used to test the model performance. Additionally, further scenarios involving sand and clay soils are examined in Appendix C, while tests on real experimental data sets are presented in Appendix B. Hydrus-1D (Simunek et al., 2013) is used to generate

Table 1
The Upper Boundary Conditions and Soil Types for the 11 Scenarios

Scenarios	S1	S2	S3	S4	S5	S6	S7	S8	S9	S10	S11
Soil type	Sandy loam	Sandy clay	Silt	Silt loam	Sandy clay loam	Clay loam	Silty clay loam		Loam		
Time (day)	Water flux density [cm day ⁻¹]										
0.25	-12	-3	-8	-8	-8	-5	-4	-12	-10	-5	-6
0.5	0.2	0.2	0	0	0	0	0.2	0	0	-5	0
0.75	0	0	0	0	0	0	0	0	0.3	0.4	0
1	0.2	0	0.5	0.4	0.4	0.5	0	0.4	0	0	0
1.25	0	0.2	0.5	0.4	0.4	0.5	0.2	0.4	0	0	-6
1.5	-10	-2	-6	-4	-4	-5	-2	-3	-10	-10	-6
1.75	0	0.1	0.5	0.4	0.4	0.5	0.1	0.4	0.3	0.4	0
2	0.2	0	0	0	0	0	0	0	0	0	0

Note. It should be noted that the value represent potential evaporation, and the upward is positive, downward is negative.

volumetric soil water content data (t, z, θ). The lower boundary is imposed with the non-flux boundary condition and the upper boundary is with atmospheric boundary condition, see Table 1 for details. Although the no-flux conditions is applied to the lower boundary at -100 cm, it does not influence the surface 20 cm soil water flow movement and the training data used in this study. Negative values represent precipitation, while positive values represent potential evaporation. The time step size is variable, with a minimum value of 0.00001 days and a maximum value of 0.001 days. The spatial mesh size is set as a constant of 0.1 cm. The training data of volumetric soil water content $\{(t_i, z_i, \theta_i)\}$ were observed at fixed depths $z = -1, -3, -5, -7, -9, -11, -13, -15, -17, -19$ cm, resulting in a total of 2010 data points. Such high-resolution observational data can be obtained using the centimeter-scale soil moisture monitoring system based on a TDR array, as proposed by Sheng et al. (2017), or through laboratory experiments using short-wave infrared (SWIR) imaging methods (Sadeghi et al., 2017). Additionally, the collocation points $\{(t_j, z_j)\}$ were generated using the Latin hypercube sampling (LHS) method. Specifically, LHS was utilized to produce a total of 10,000 collocation points, which were integral to the robustness of our training process. A total of 11 scenarios are considered, eight of which are used to investigate the performance of PINN-AWL under different soil types, and three are used to study its performance under different upper boundaries of the same soil, which is loam in this study. The van Genuchten-Mualem model (Mualem, 1976; van Genuchten, 1980) is applied to describe the soil water characteristic curve. The soil hydraulic parameters of van Genuchten-Mualem model of eight soils are shown in Table 2.

All experiments are performed on a single NVIDIA 2080ti GPU or single NVIDIA 4090 GPU. The network weights are initialized using the Xavier normal distribution (Glorot & Bengio, 2010), and all biases are set to zero. For optimization, the Adam optimizer with 6,000 epochs followed by the LBFGS optimizer with 3,000 epochs is used to train the network. The training process will terminate when the absolute difference between the loss values

of two adjacent iteration steps is less than a certain threshold value (we choose $\epsilon = 10^{-7}$). Although the number of iterations appears small, all network models completed training within 7,200 iterations. Training was terminated when the difference in loss between two iterations was less than 10^{-7} , and all networks were adequately trained.

Table 2
The Soil Hydraulic Parameters of the van Genuchten-Mualem Model for the Eight Types of Soils (Carsel & Parrish, 1988)

	θ_r [—]	θ_s [—]	α [cm ⁻¹]	n [—]	K_s [cm day ⁻¹]	l [—]
Sandy loam	0.065	0.41	0.075	1.89	106.1	0.5
Loam	0.078	0.43	0.036	1.56	24.96	0.5
Silt	0.034	0.46	0.016	1.37	6	0.5
Silt loam	0.067	0.45	0.02	1.41	10.8	0.5
Sandyclay loam	0.1	0.39	0.059	1.48	31.44	0.5
Clay loam	0.095	0.41	0.019	1.31	6.24	0.5
Silty clay loam	0.089	0.43	0.01	1.23	1.68	0.5
Sandy clay	0.1	0.38	0.027	1.23	2.88	0.5

5. Performance Evaluation of the PINN-AWL

The hyper-parameters ($\alpha_r, \beta_r, \alpha_\theta, \beta_\theta$) in Equations 12 and 13 used for calculating the weights were set as (10, 1, 100, 1), and the hyper-parameters ($\lambda_\theta, \lambda_r$) in Equation 14 were set as (10, 1), if not stated otherwise.

5.1. With Different Network Architectures

To simply investigate the effect of network's architecture on the performance of PINN-AWL, we made two assumptions: (a) in the three sub-networks

$\hat{\psi}(t, z)$, $\hat{K}(\hat{\psi})$ and $\hat{\theta}(\hat{\psi})$, the number of units in each hidden layer is the same and remains consistent; (b) the two sub-networks $\hat{K}(\hat{\psi})$ and $\hat{\theta}(\hat{\psi})$ have the same network structure, but do not share parameters. The combination of the number of hidden layers (1, 2, 3, 4, 5, 6) and the number of neurons per layer (10, 20, 40, 80) is studied, and it will result in 144 ($4 \times 6 \times 6$) different network architectures based on the aforementioned assumptions. Herein, only the results of the soil type of loam, which is scenario eight in Table 1, are listed in detail for illustration in Figure 5, including the relative L_2 errors of volumetric soil water content ϵ^θ , soil matric potential ϵ^ψ , unsaturated hydraulic conductivity ϵ^K , soil water flux density ϵ^q under different network architectures. It can be found that: (a) ϵ^θ are smaller than 0.1 for most network architectures; (b) considering the accuracy of the three other quantities, small errors can be obtained for the subnetwork $\hat{K}(\hat{\psi})$ and $\hat{\theta}(\hat{\psi})$ with fewer hidden layers; (c) in many cases, increasing the number of neurons per hidden layer from 10 to 40 significantly decreases errors, while continuing to increase will not result in much improvement; (d) for all the four prediction measurements, the PINN-AWL consisting of $\hat{\psi}(t, z)$ with 6 hidden layers and 40 neurons per layer, followed by $\hat{K}(\hat{\psi})$ and $\hat{\theta}(\hat{\psi})$ with 3 hidden layers and 40 neurons per layer, achieves almost the best performance, as shown in Figures 5c, 5g, 5k, and 5o with red marks and Figures 6a–6c. Thus this network structure will be used by default in all remaining tests.

It should be noted that ϵ^ψ is very large while ϵ^K and ϵ^q are relatively small for some network architectures. For example, with the network consisting of $\hat{\psi}(t, z)$ with four hidden layers followed by $\hat{K}(\hat{\psi})$ and $\hat{\theta}(\hat{\psi})$ with 1 hidden layer (10 neurons per layer), ϵ^ψ is 4×10^5 , but ϵ^K and ϵ^q are 0.112 and 0.080, respectively, as shown in Figures 5a, 5c, 5i, and 5m with blue marks. The predicted soil matric potential $\hat{\psi}$, unsaturated hydraulic conductivity $\hat{K}$, soil water flux density $\hat{q}$ are shown in Figures 6d–6f, from which we see that although the network fits the soil water flux density very well, it still fails to capture the current water retention curve, indicating that ϵ^q is not enough to reflect the prediction accuracy of the model. We think the reason is the soil water flux density is a product $\hat{K}$ and $\partial\hat{\psi}/\partial z + 1$, which does not mean that the prediction of matric potential or unsaturated hydraulic conductivity is accurate. Thus, a comprehensive comparison of all four prediction measurements are indeed important.

5.2. With Different Types of Soils

The proposed model PIN-AWL is mainly based on volumetric soil water content data to simulate soil water flow and invert the constitutive relationship without knowing the initial and boundary conditions, but cannot be used for forecasting volumetric water content under different initial and boundary conditions. The obtained constitutive relationships applicable within numerical and neural network models that necessitate such a constitutive relationship. We now examine the performance of PINN-AWL on the eight different types of soils (scenarios S1–S8) in Table 1. The comparison of the predicted volumetric soil water content by the PINN-AWL and the reference values at $t = 0.5, 1, 1.5$ d are shown in Figure 7, from which we see that the predicted curve fits the true values very well. Table 3 further shows the relative L_2 errors of volumetric soil water content ϵ^θ , which is smaller than 0.006 for all scenarios, indicating that the predictions are quite accurate.

The comparisons between the water retention curves predicted by the PINN-AWL and those from the van Genuchten-Mualem model for the 8 types of soils are shown in Figure 8, indicating that they matched very well. We remark that the PINN-AWL can predict the water retention curves at both low and high volumetric soil water contents in all 8 scenarios, while it is hard for the monotonic PINN to catch the correct solution due to the stronger nonlinearity at the two ends of the curve and the scarcity of data existing on those regions. The satisfactory results produced by the proposed PINN-AWL mostly attribute to the adaptively weighted loss function, which makes the network gradually focus more on regions with significant losses during training. The comparisons between the unsaturated hydraulic conductivity predicted by the PINN-AWL and those from the van Genuchten-Mualem model for the eight types of soils are shown in Figure 9, indicating that they are also well estimated although there are no additional constraints in the network. It also can be seen that PINN-AWL can predict the unsaturated hydraulic conductivity well by minimizing the residual of Richards' equation. Due to the fact that the predicted saturated soil water content of sandy loam (S1) is less than the true value (see Figure 8a), the predicted hydraulic conductivity function is absent at the wet end, (see Figure 9a). In addition, PINN-AWL performs well in predicting the hydraulic conductivity both in the wet and dry ends based on the results of Figure 9. Our training time are 40–60 min on an NVIDIA 2080Ti GPU and fewer than 10 min on NVIDIA 4090 GPU. The exact training time depends on the specific task. Hydrus-1D completed the inversion within 1 min under the simple homogeneous conditions when using the same data set. Despite the longer training time, PINN-AWL can effectively handle

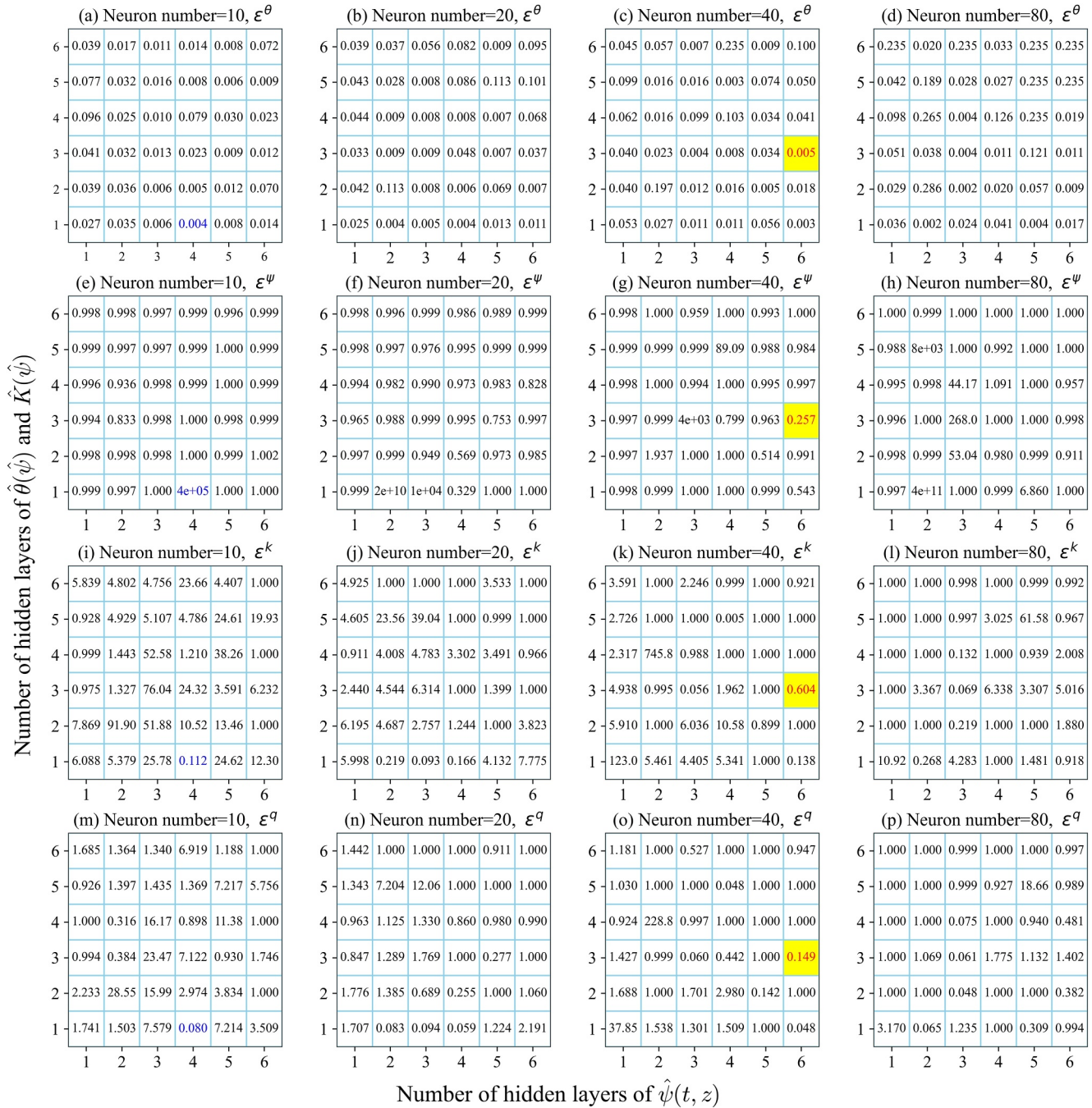


Figure 5. Comparison of the relative L_2 errors of volumetric soil water content ϵ^θ , the soil matric potential ϵ^ψ , the unsaturated hydraulic conductivity ϵ^K , and the soil water flux density ϵ^q under various PINN-AWL network architectures with varying neuron numbers and hidden layer configurations.

unknown boundary conditions, a significant advantage over Hydrus-1D. Besides, the results indicate that a single set of hyperparameters demonstrates robust performance across a variety of scenarios, suggesting their suitability for a broad range of applications.

5.3. Under Different Boundary Conditions

We test the performance of PINN-AWL under three different upper boundary conditions, scenarios S9–S11. The comparisons between the volumetric soil water content predicted by the PINN-AWL and the true values at $t = 0.5$,

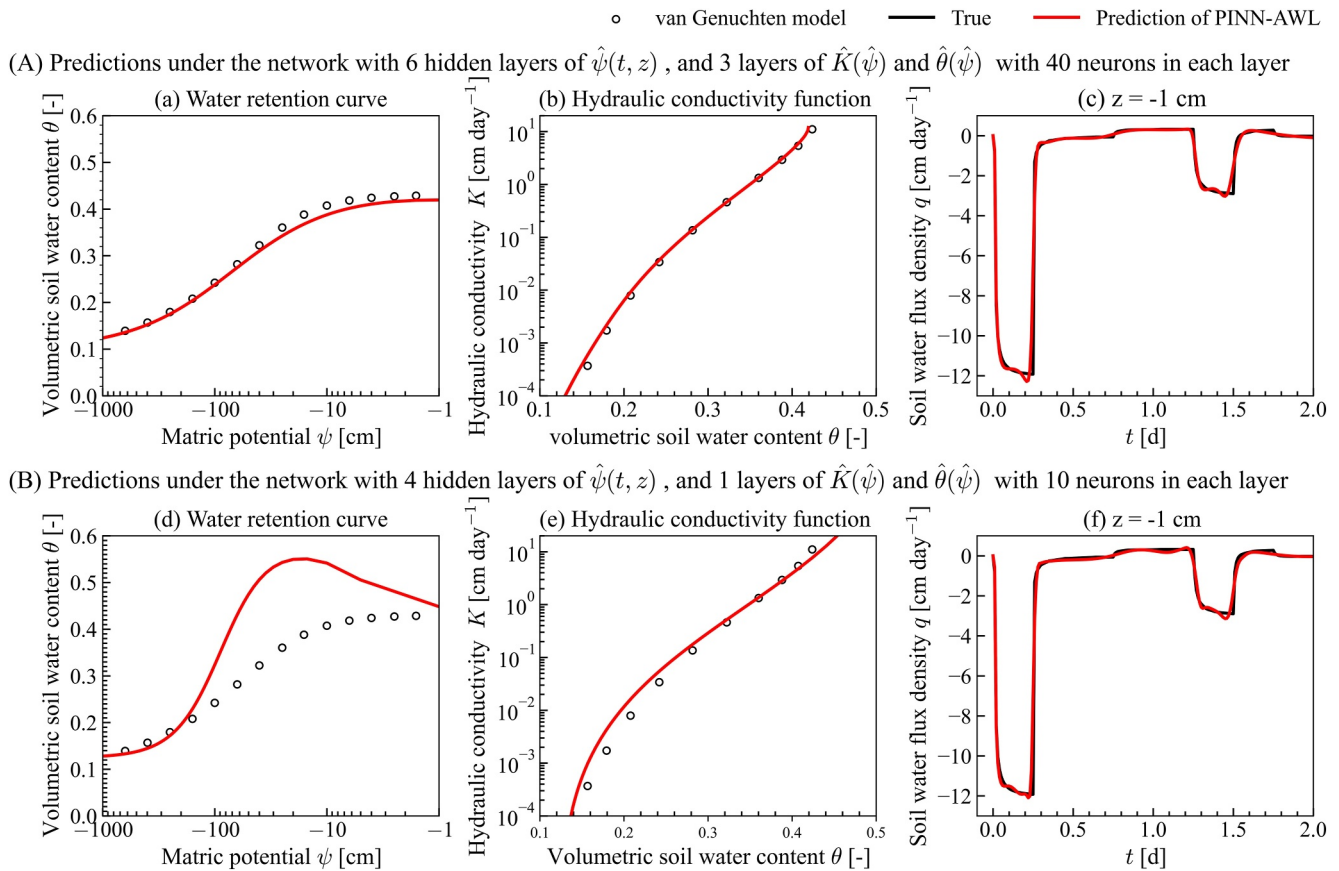


Figure 6. Comparisons of constitutive relationship from the van Genuchten-Mualem model and the ground truth of soil water flux density at $z = -1$ cm with the ones predicted by the PINN-AWL under two different networks' architecture.

1, 1.5 d are plotted in Figure 10. The relative L_2 errors under the three upper boundary conditions are shown in Table 1. All the errors were smaller than 0.003. In terms of water retention curves and hydraulic conductivity functions, they are constitutive relationships for soil water flow problems and do not depend on the setting of boundary conditions. The results in Figure 11 confirm this point, where the predicted values under the three boundary conditions are always consistent with those from the van Genuchten-Mualem model, indicating the robustness of PINN-AWL in inverting the constitutive relationships of Richards' equation.

5.4. With Different Number of Collocation Points

In this section, we explore the impact of the number of collocation points on the performance of PINN-AWL. Collocation points are not only the carrier of the physical constraints that the network needs to meet in the training process, but also the key factor to ensure the accuracy and generalization ability of the model. Through the appropriate collocation points strategy, the performance of PINNs in solving PDEs can be significantly improved (Hou et al., 2023). We investigate the performance of PINN-AWL with different numbers of collocation points, specifically at 100, 1,000, 10,000, and 100,000, to understand how this factor influences the model's effectiveness. Figure 12 shows the prediction of volumetric water content by PINN-AWL at $t = 0.5, 1.5$ d for different numbers of sampling points. As illustrated in Figure 12, we can observe that when the number of collocation points is 100, there is a noticeable discrepancy between the predicted curve of volumetric soil water content and the data points. For 1,000, 10,000, and 100,000 collocation points, the predicted curve fits the data point very well. Figure 13 depicts the predictive performance of PINN-AWL regarding constitutive relations across varying numbers of collocation points. It is evident that the PINN-AWL achieves accurate prediction of the constitutive relationship only with a sufficient quantity of collocation points—specifically, when the count exceeds 10,000. This observation suggests that the insufficiency number of collocation points hinders the network's capacity to thoroughly assimilate the

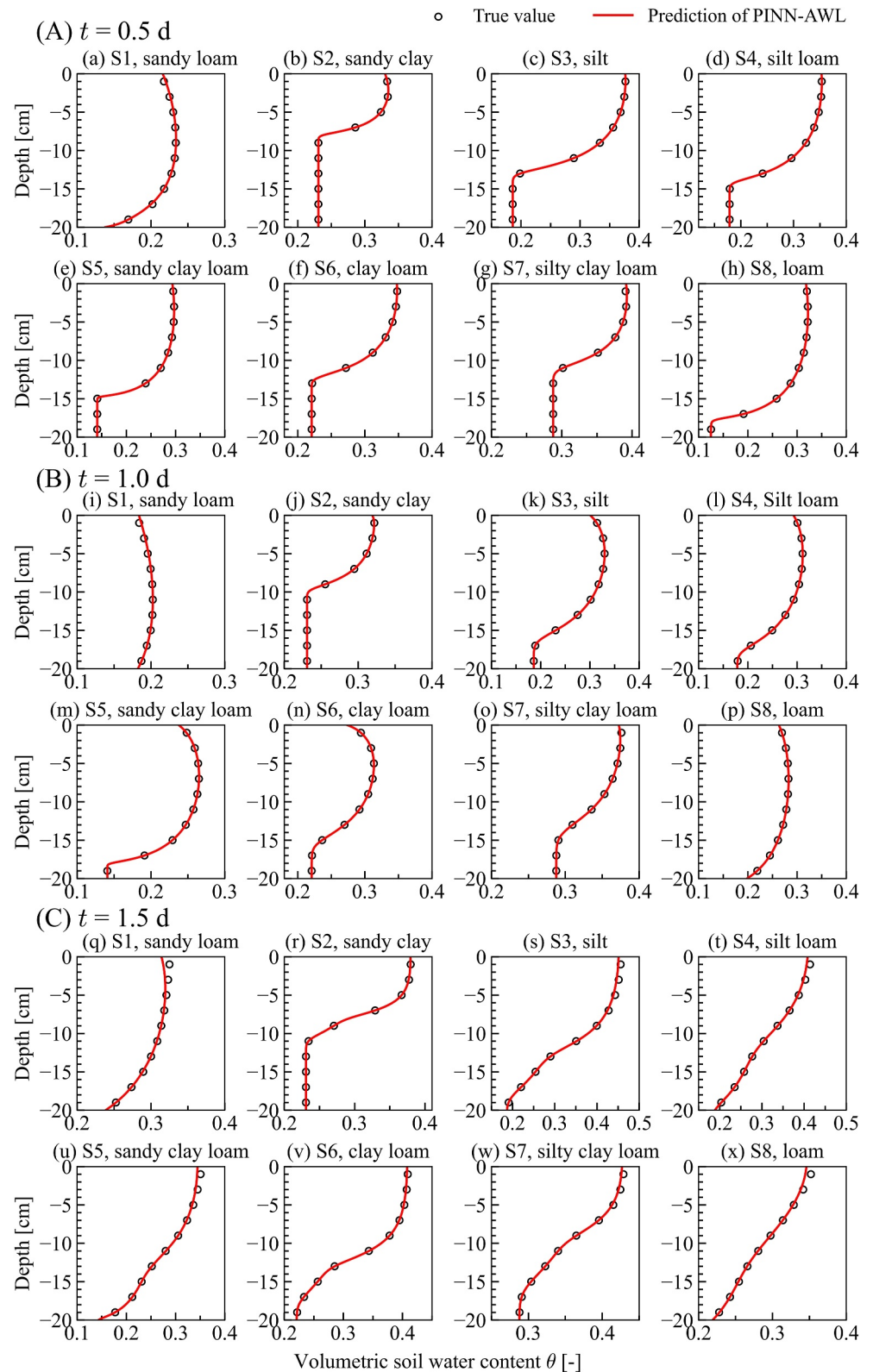


Figure 7. Comparisons of predicted volumetric soil water content θ by PINN-AWL and the true values for the 8 types of soils (scenarios S1–S8) at different times: (A) $t = 0.5$ d; (B) $t = 1.0$ d; (C) $t = 1.5$ d.

Table 3

The Relative L_2 Errors of the Volumetric Soil Water Content ϵ^θ

Scenario	S1	S2	S3	S4	S5	S6	S7	S8	S9	S10	S11
Soil type	Sandy loam	Sandy clay	Silt	Silt loam	Sandy clay loam	Clay loam	Silty clay loam	Loam			
PINN-AWL	0.0054	0.0024	0.0038	0.0033	0.0051	0.0035	0.0019	0.0043	0.0027	0.0025	0.0054
Monotonic PINN	0.0101	0.0068	0.0111	0.0187	0.0081	0.0105	0.0085	0.0089	—	—	—
Standard PINN	0.0131	0.0440	0.0038	0.0036	0.0063	0.0056	0.0265	0.0202	—	—	—

embedded physical constraints, consequently failing to precisely mirror the actual physical process. Utilizing a substantial number of collocation points can lead to a significant increase in computational resource expenditure. Consequently, a recommended number of collocation points is 10,000 in this study.

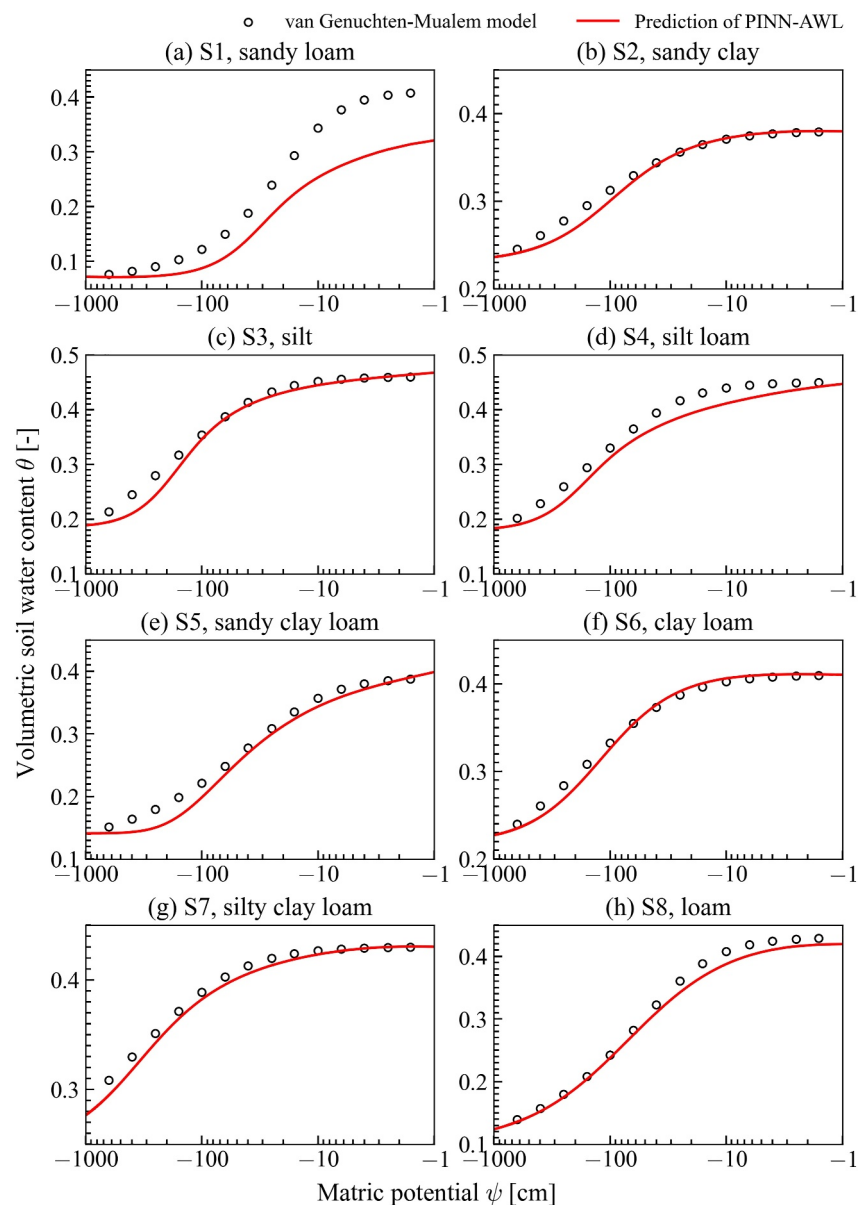


Figure 8. Comparisons between the water retention curves predicted by PINN-AWL and those from the van Genuchten-Mualem model for the eight types of soils (scenarios S1–S8).

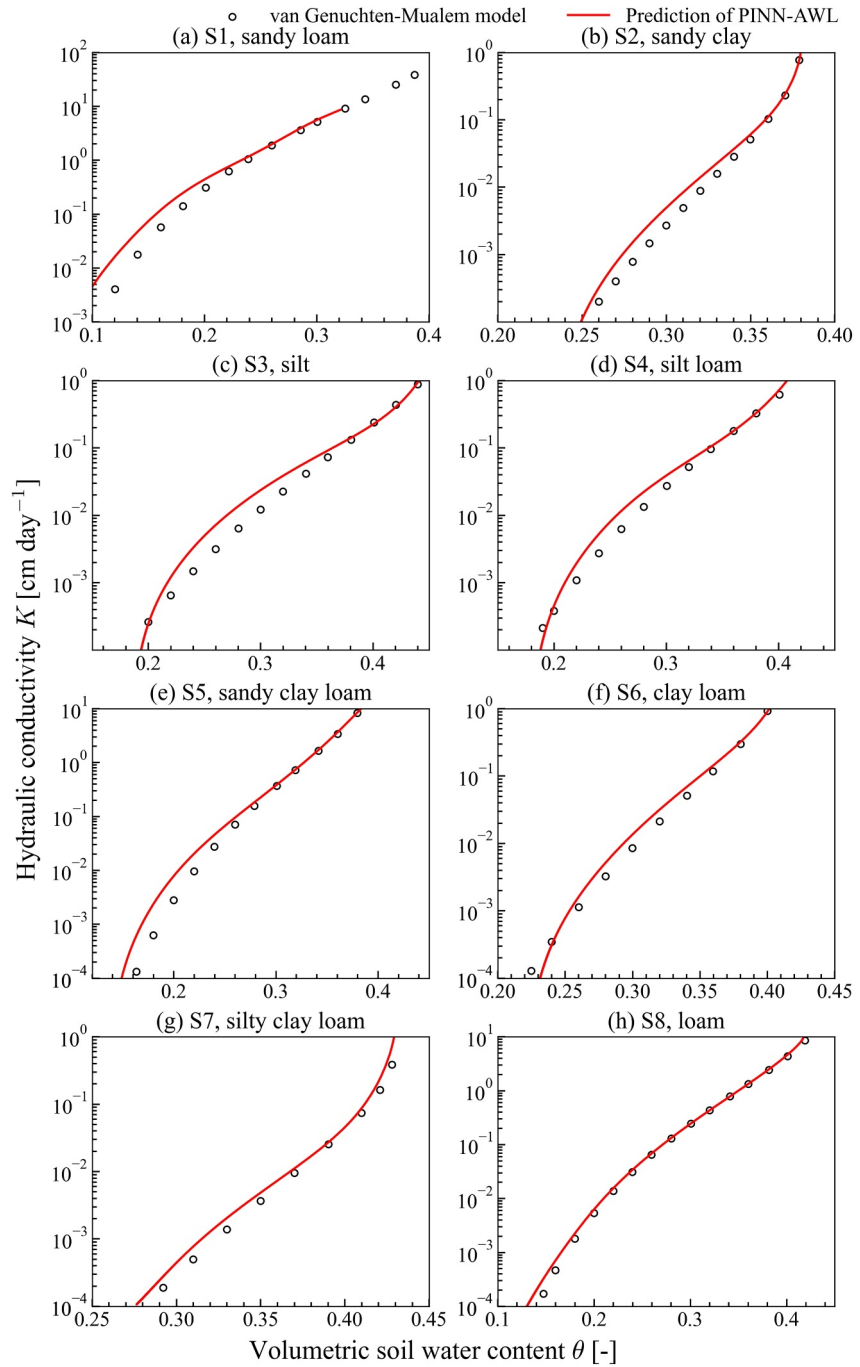


Figure 9. Comparisons between the hydraulic conductivity K predicted by PINN-AWL and those from the van Genuchten-Mualem model for the eight types of soils (scenarios S1–S8).

6. Ablation Study on the Adaptively Weighted Loss

In this section, we conduct a comprehensive comparison on the performance of the standard PINN, the monotonic PINN, and the PINN-AWL, and investigate the influence of the adaptive weights $(\alpha_r, \beta_r, \alpha_\theta, \beta_\theta)$ and the balancing weights $(\lambda_\theta, \lambda_r)$ on PINN-AWL.

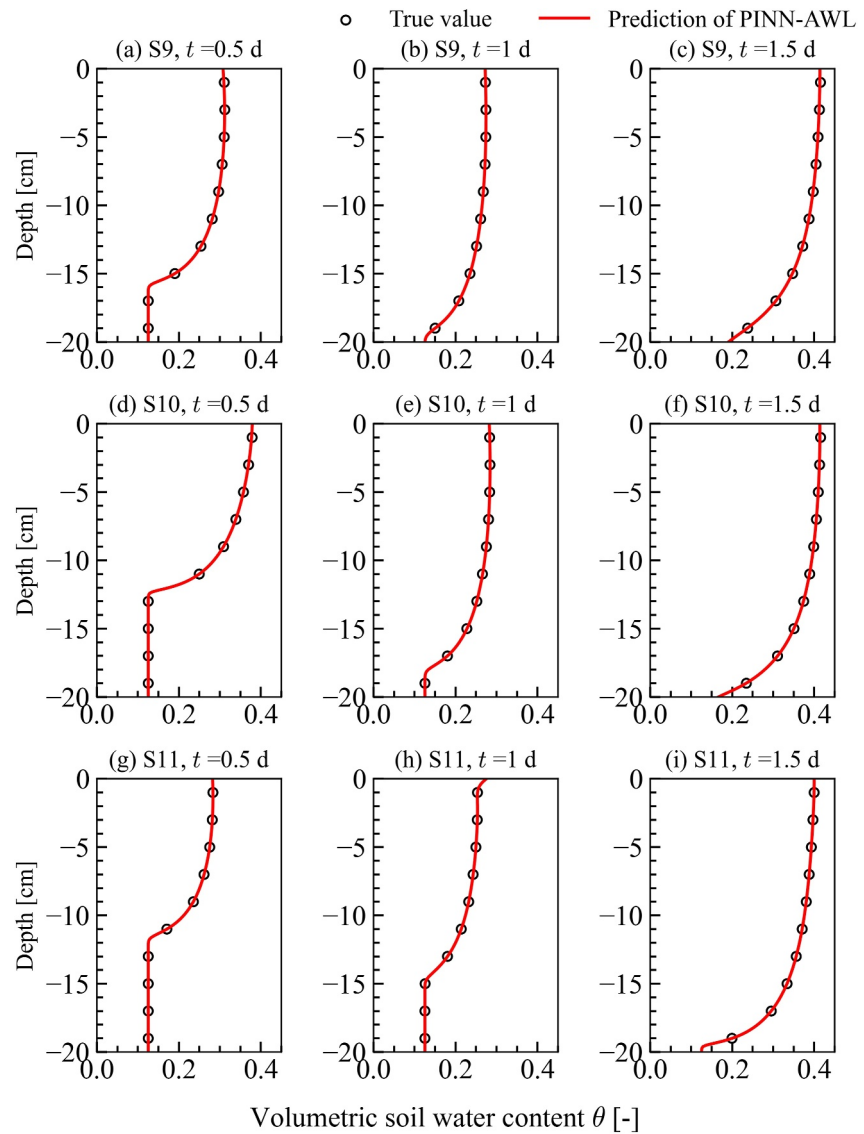


Figure 10. Comparisons of volumetric soil water content predicted by PINN-AWL and the true values for loam under three boundary conditions (S9–S11) at $t = 0.5, 1.0,$ and 1.5 d: (a–c) S9, (d–f) S10, (g–i) S11, with columns corresponding to $t = 0.5, 1.0,$ and 1.5 d, respectively.

6.1. Comparison of the PINN-AWL With the Standard and Monotonic PINNs

We test the performance of the standard PINN, the monotonic PINN, and the PINN-AWL on the eight different types of soils (scenarios S1–S8). Although the three PINNs share an identical network architecture, each has demonstrated commendable performance within this framework. This consistency allows for an equitable comparison of their results under the same structural conditions. The predicted volumetric soil water content from the models at $t = 0.5, 1, 1.5$ d are shown in Figure 14, and their relative L_2 errors are summarized in Table 3. The relative L_2 errors of the three PINN models are all in an acceptable range, with values less than 0.03. However, obvious overfitting and deviations can be found in those abrupt change points or upper boundary for the standard PINN as shown in Figures 14g, 14j, 14r, and 14w, and underfitting phenomenon can be found adjacent to the upper boundary for the monotonic PINN as shown in Figures 14q, 14t, and 14w, while the PINN-AWL has the smoothest prediction curves for all the 8 types of soils. The performance of the soil water flux density from the three models can be found in Figure 15, from which we observe: (a) the PINN-AWL can accurately estimate the soil water flux density for all the 8 types of soils; (b) the standard PINN fails to predict the soil water flux density in some scenarios, such as sandy clay (S2), silty clay loam (S7) and loam (S8); (c) the monotonic PINN only

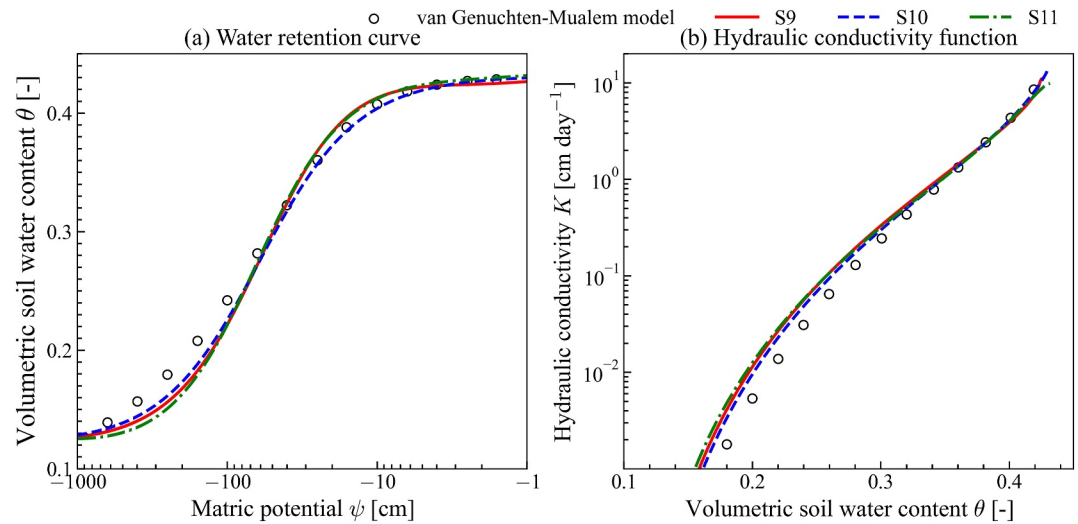


Figure 11. Comparisons of the constitutive relationship: (a) water retention curve and (b) hydraulic conductivity function, predicted by the PINN-AWL and those from the van Genuchten-Mualem model under the three different boundary conditions (scenarios S9–S11).

provides a rough prediction of the soil water flux density because it fails to capture the maximum value and sharp changes of the soil water flux density. The standard PINN uses the MSE of each point in its loss function and can not capture subtle details in those regions with abrupt change. The proposed PINN-AWL specifies the contribution of the mean absolute error (MAE) of each point to the loss function by multiplying adaptive weights, which helps to focus training on regions with larger errors. This could be the main reason for the success of PINN-AWL.

The comparisons of the constitutive relationships, including the water retention curves and the unsaturated hydraulic conductivity functions, predicted by the three PINN models and those from the van Genuchten-Mualem model are shown in Figure 16. Obviously, the PINN-AWL outperforms the other two PINN models. The standard PINN can inverse the water retention curve and the hydraulic conductivity function successfully for the four types of soils, for example, silt (S3), silt loam (S4), sandy clay loam (S5) and clay loam (S6), it outperforms than Monotonic PINN in these four types of soils, and it outperforms than PINN-AWL in silt (S3), silt loam (S4), sandy clay loam (S5). While it fails for the other four soil types. The reason may be due to the nonlinearity of the water retention curve or the distribution of the training data used. From Figure 1, we can find out the parameter “ n ,” located on the exponent, controls the nonlinearity of the WRC; this is also true for HCF. The standard PINN performs well for soils with moderate nonlinearity, characterized by a parameter “ n ” ranging from 1.31 to 1.47.

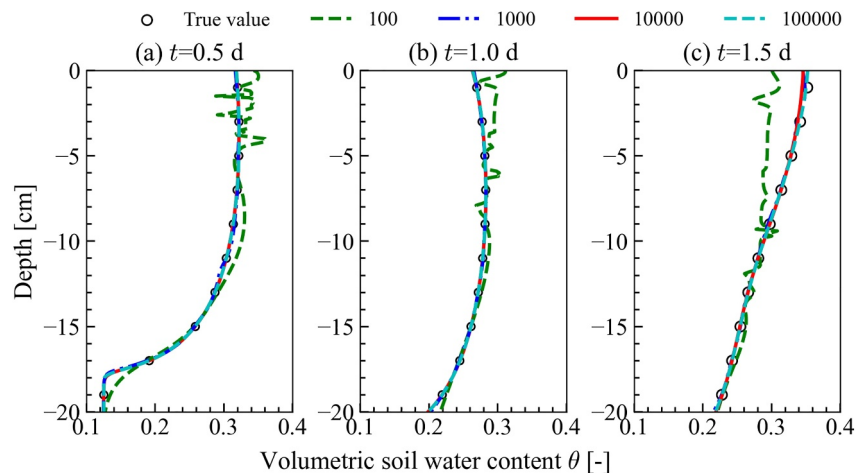


Figure 12. Comparisons of volumetric soil water content predicted by PINN-AWL with different number of collocation points at times (a) $t = 0.5$ d, (b) $t = 1.0$ d, and (c) $t = 1.5$ d.

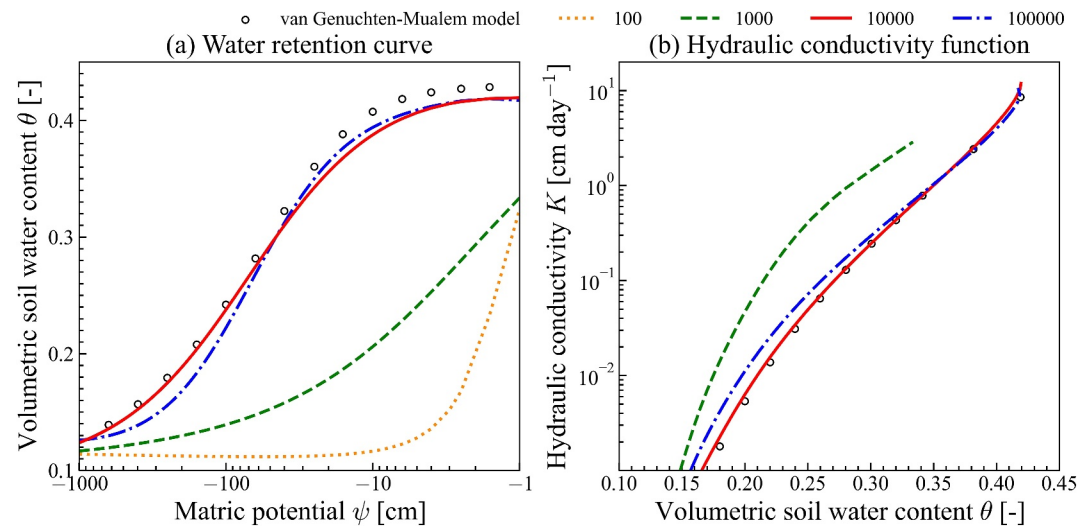


Figure 13. Comparisons of the constitutive relationship: (a) water retention curve and (b) hydraulic conductivity function, from the van Genuchten-Mualem model with those predicted by PINN-AWL with different number of collocation points.

However, its predictive accuracy declines for soils exhibiting either strong nonlinearity, such as sandy loam ($n = 1.89$) and loam ($n = 1.56$), or weak nonlinearity, such as silty clay loam and sandy clay, both with $n = 1.23$. One possible reason for this is that standard PINNs are suitable for capturing moderately nonlinear features. For more complex or simpler nonlinear features, the standard PINN may lack the necessary flexibility. The PINN-AWL model places greater emphasis on capturing the wet end of the soil moisture so that it does not perform as well as standard PINN in dry conditions in silt (S3), silt loam (S4). This pattern is evident in the case of sandy clay loam, where PINN-AWL shows improved predictions at the wet end compared to the standard PINN, yet underperforms than Standard PINN at the dry end. While the standard PINN delivers satisfactory outcomes in certain cases, PINN-AWL demonstrates greater robustness. The adaptive learning mechanism within the PINN-AWL framework ensures that it can effectively invert the constitutive relationships of soils, regardless of their position within the range of “ n ” values. The monotonic PINN becomes more robust and stable in training by adding the monotonic constraints into the two subnetworks $\hat{K}(\psi)$ and $\hat{\theta}(\psi)$, and achieves better prediction results than the standard PINN. In the scenarios of sandy loam (S1), sandy clay (S2) and sandy clay loam (S5), the predicted water retention curves are very close to the true curves when the matric potential ranged from $-1,000$ to -100 cm, thanks to the monotonicity of the curves. However, it is also due to this strong monotonicity constraint that large deviations occur in other parts, which are not the true characteristics of the curve. These results show that the PINN-AWL can effectively grasp the nonlinear variations of water retention curves and hydraulic conductivity functions to achieve more robust and accurate predictions.

To clarify the effect of increasing training iterations, we conducted an extended analysis on both silt and sandy clay. Our results show that while the Standard PINN benefits from more iterations in silt—achieving similar inversion accuracy at both 6,000 and 100,000 iterations—the Monotonic PINN exhibits consistent but slightly less accurate performance in this case. In sandy clay, increasing iterations beyond 6,000 does not yield improved results for either method. In fact, for the Standard PINN, excessive iterations may lead to overfitting, resulting in degraded performance. These findings suggest that while longer training may offer some improvements in certain cases, our results indicate that using 6,000 iterations, together with an appropriate optimizer and learning rate, is generally adequate to achieve stable performance and avoid overfitting.

Additionally, different initialization schemes (e.g., varying random seeds) exert distinct impacts on network performance. The standard PINN exhibits significant variability in results across different seeds, while Monotonic PINN and PINN-AWL maintain greater stability due to monotonicity constraints and adaptive weighting strategies, respectively. Notably, the initialized seeds for PINN-AWL own strong generalization across different soil types, as demonstrated in the case studies of this work, further confirming its robustness. These observations illustrate the necessity of extensive ablation experiments to identify initialization seed values that enhance the robustness of PINN models.

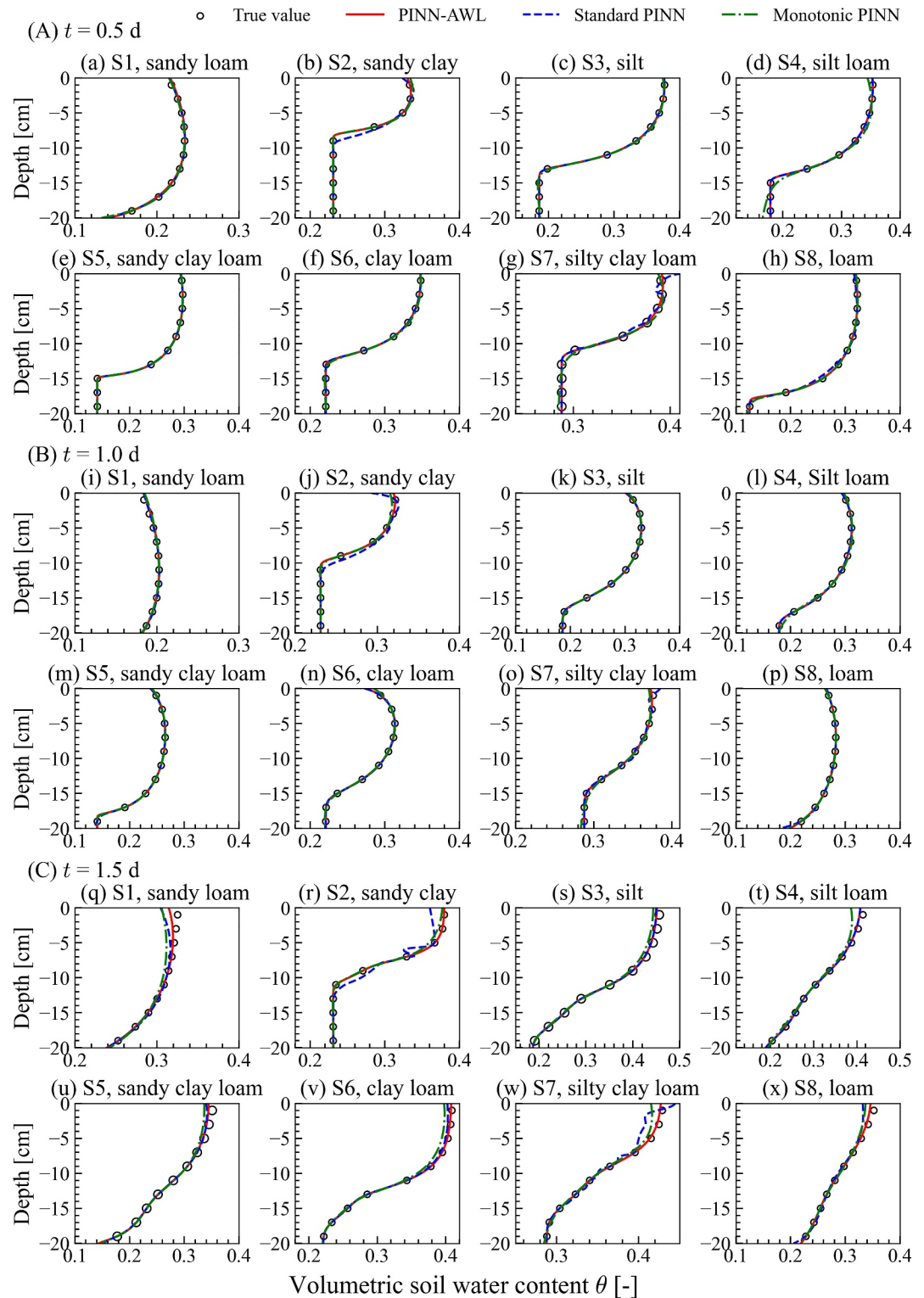


Figure 14. Comparisons of predicted volumetric soil water content by the three PINNs (PINN-AWL, Monotonic PINN and Standard PINN) and the true values for the 8 types of soils at $t = 0.5, 1, 1.5$ d.

Both the Monotonicity and adaptively weighted loss method contribute to enhancing the prediction outcomes to a certain degree, introducing monotonicity constraints into the PINN-AWL may improve prediction results. Thus, we study the performance of Monotonic PINN-AWL in the loam scenario (S8). Visual representations of the

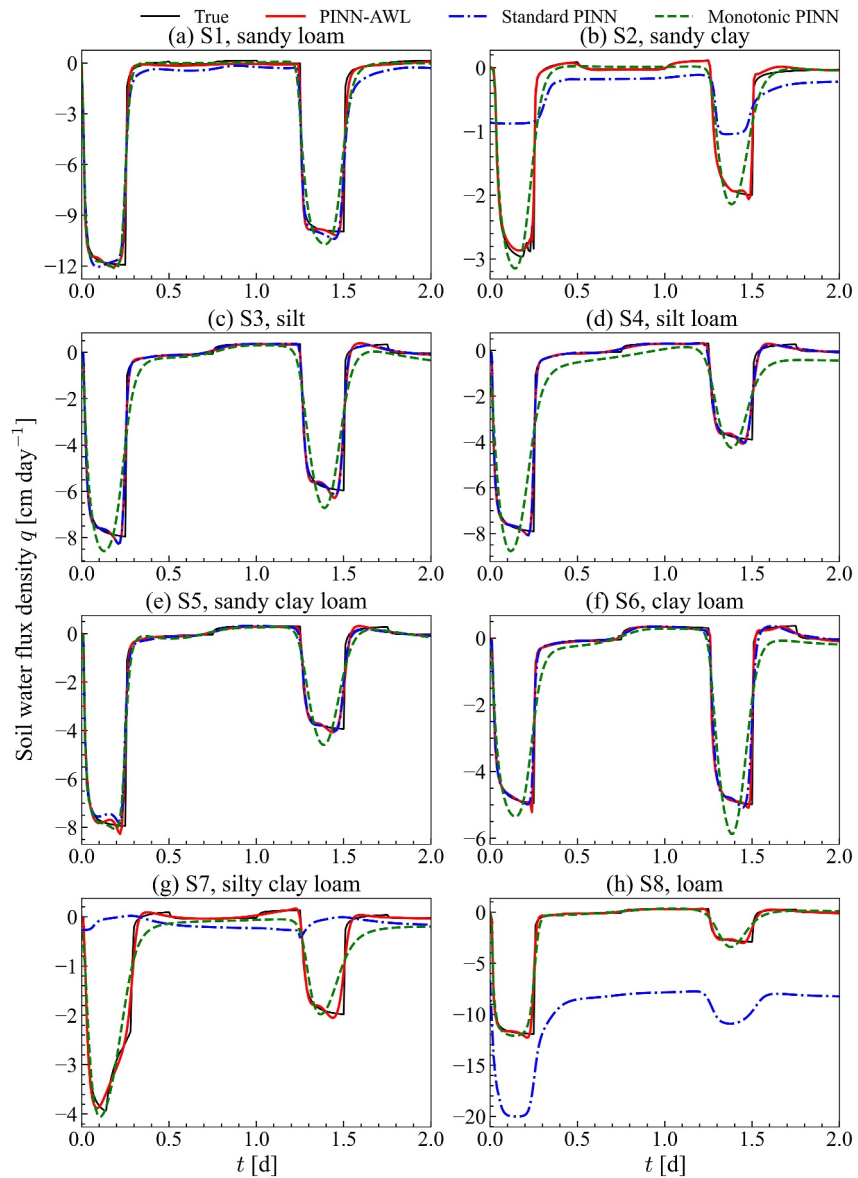


Figure 15. Comparisons of predicted soil water flux density by the three PINNs (PINN-AWL, Monotonic PINN and Standard PINN) and the true values for the eight types of soils at $z = -1$ cm: (a) S1, sandy loam; (b) S2, sandy clay; (c) S3, silt; (d) S4, silt loam; (e) S5, sandy clay loam; (f) S6, clay loam; (g) S7, silty clay loam; (h) S8, loam.

volumetric soil water content and the constitutive relationship, as predicted by the Monotonic PINN-AWL, are displayed in Figures 17 and 18, respectively. It becomes apparent that the incorporation of monotonicity into the PINN-AWL does not yield a significant improvement in the accuracy of the constitutive relationship predictions, the results are very similar to those predicted by Monotonic PINN as shown in Figure 16. It suggests that adaptively weighted loss methods are sufficient in capturing the physical essence, the additional imposition of monotonicity may not be necessary. Instead, it may lead to an overconstraint of the model, which could potentially diminishing the effectiveness of the adaptively weighted loss approach. In conclusion, we recognize the potential benefits of using an adaptively weighted loss in conjunction with monotonicity constraints, our experimental results indicate that the current implementation of monotonicity may not lead to improved performance. We believe that the proposed PINN-AWL provide a robust and effective solution for the problems considered in our study.

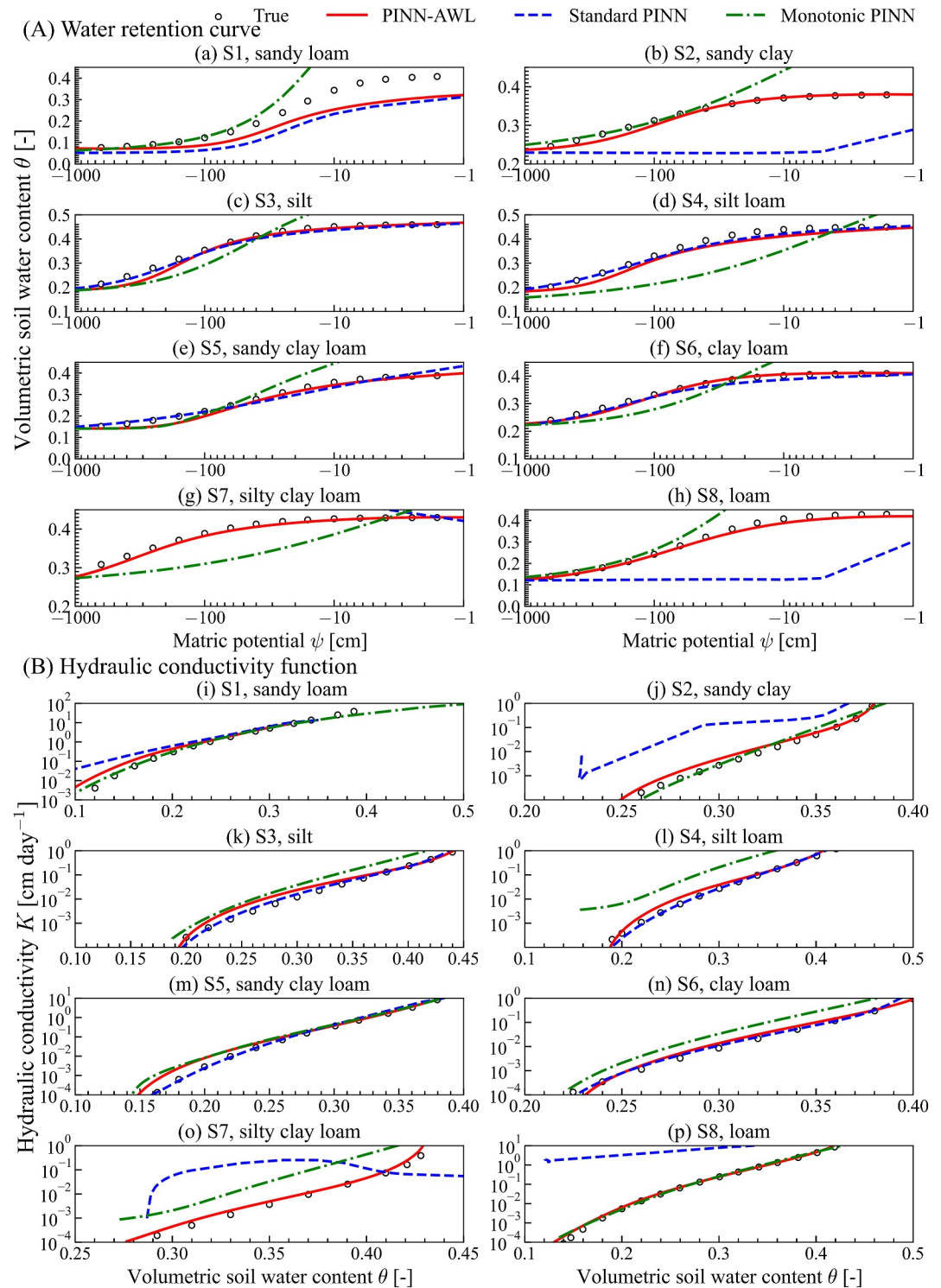


Figure 16. Comparisons of the constitutive relationship: (A) water retention curve and (B) hydraulic conductivity function, from the van Genuchten-Mualem model with those predicted by the three PINNs (PINN-AWL, Monotonic PINN and Standard PINN).

6.2. Influence of Hyper-Parameters (α_r , β_r , α_θ , β_θ) in Calculating Adaptive Weights

Two types of hyper-parameters are used to adjust the adaptive weights defined in Equations 12 and 13, where α and β determine the range and the minimum value of the weights, respectively, resulting in weights varying from α to

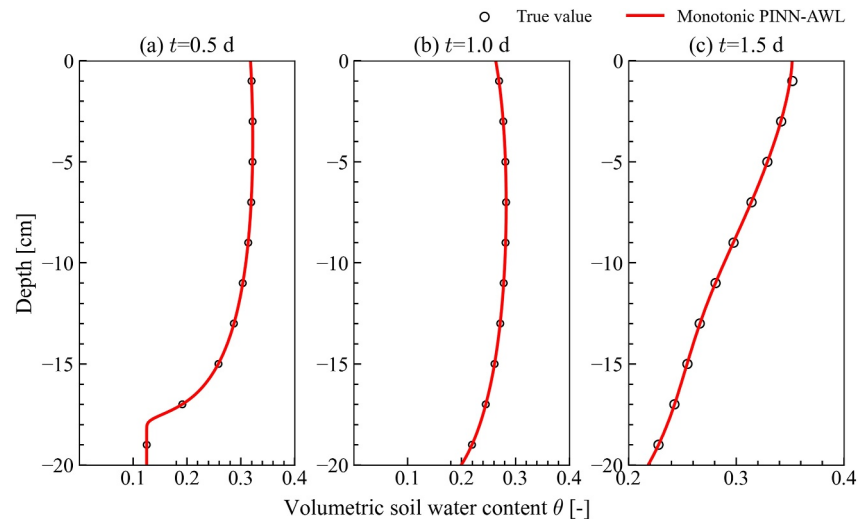


Figure 17. Comparisons of volumetric soil water content predicted by Monotonic PINN-AWL and the true values at different times: (a) $t = 0.5$ d; (b) $t = 1.0$ d; (c) $t = 1.5$ d.

$\alpha + \beta$. We set β to be a constant of 1, and let α take values between 1, 10 and 100 to form three groups. Since the loss function consists of two items, including the residual of Richards' equation and the loss of volumetric soil water content data, there are totally nine groups of hyper-parameters combinations (G1-G9) as shown in Table 4. The soil type of loam (S8) is used herein as the illustration example. Other settings are the same as mentioned above.

The volumetric soil water content predicted by the PINN-AWL with these nine different groups of $(\alpha_r, \beta_r, \alpha_\theta, \beta_\theta)$ at $t = 0.5, 1.0$, and 1.5 d are plotted in Figure 19. While the relative L_2 errors of the volumetric soil water content are shown in Table 5. It can be found that the relative L_2 errors under some groups were large, for example, G7 and G8. In these cases, the fact that α_r is larger than α_θ makes the network pay more attention to the constraints of satisfying Richards' equation while decreases the effects of data. On the contrary, when α_r is significantly greater than α_θ (e.g., G3), it can also lead to inaccurate results, where overfitting occurs in the top soil layer as shown in Figure 19. Moreover, there are significant deviations in the upper boundary at $t = 1, 1.5$ d for G, as shown by the bright green solid line in Figure 19c. Other groups, including G2, G4-G6, and G9, can obtain accurate results when considering the prediction of volumetric soil water content.

It is more difficult for PINN-AWL to obtain reasonable constitutive relationships, and the corresponding results are shown in Figure 20, from which we see that only G4 and G6 can obtain both the accurate water retention curve

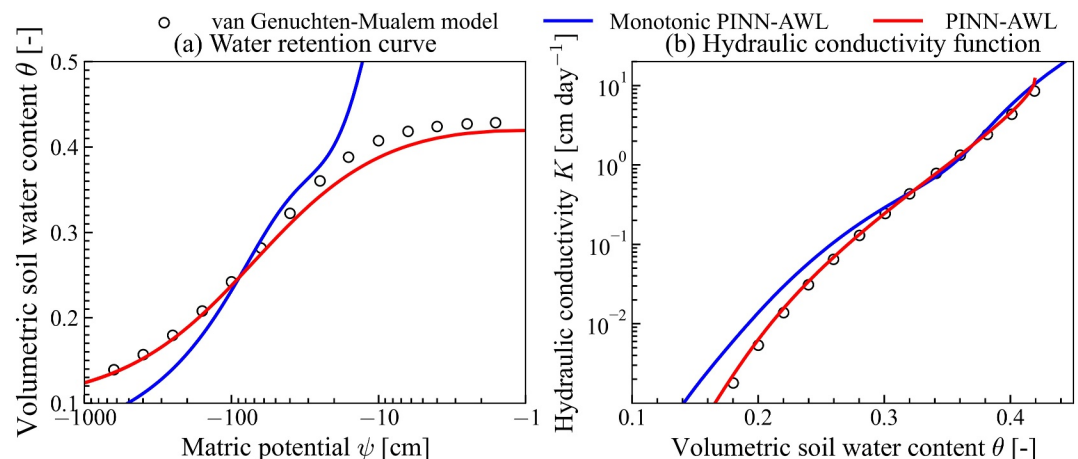


Figure 18. Comparisons of the constitutive relationship: (a) water retention curve and (b) hydraulic conductivity function, from the van Genuchten-Mualem model with those predicted by Monotonic PINN-AWL and PINN-AWL.

Table 4
Different Combinations of $(\alpha_r, \beta_r, \alpha_\theta, \beta_\theta)$ in Calculating Adaptive Weights,
Where $\beta_\theta = \beta_r = 1$

Group	α_r	α_θ	Group	α_r	α_θ	Group	α_r	α_θ
G1	1	1	G4	10	1	G7	100	1
G2	1	10	G5	10	10	G8	100	10
G3	1	100	G6	10	100	G9	100	100

and hydraulic conductivity function, indicating relatively large value of α_r is required for training. Since there is no measured matric potential data used in two subnetworks $\hat{K}(\hat{\psi})$ and $\hat{\theta}(\hat{\psi})$, learning the accurate Richards' equation is more important for obtaining the fitted water retention curve and hydraulic conductivity function.

6.3. Influence of the Hyper-Parameters $(\lambda_\theta, \lambda_r)$ in Calculating the Loss Function

Two hyper-parameters λ_θ and λ_r are used to balance the two loss terms of $\mathcal{L}_{AW}$ defined in Equation 14, and we now investigate their influence on the performance of PINN-AWL through nine different groups of combinations $(\lambda_\theta, \lambda_r)$ given in Table 6, marked as W1-W6. The relative L_2 errors and predicted curves of volumetric soil water content under the six groups are shown in Table 7 and Figure 21, respectively. The predicted water retention curve and the hydraulic conductivity function under the six groups are shown in Figure 22.

The loss function $\mathcal{L}_{AW}$ with W1 only contains the information of volumetric soil water content data, while ignoring the constraint of Richards' equation, and in this case, although the relative L_2 error is acceptable (see Table 7), the PINN-AWL fails to inverse the water retention curve and the unsaturated hydraulic conductivity function (see Figure 22). In contrast, since the loss function $\mathcal{L}_{AW}$ with W2 only contains the information of Richards' equation, PINN-AWL not only fails predict the volumetric soil water content (see Table 7), but also the predicted curve of the volumetric soil water content becomes a vertical line (see Figure 21), which is completely wrong. We also find from Table 7 that when λ_r fixed as 1, the relative L_2 error of the volumetric soil water content gradually decreases with the increasing of λ_θ until it reaches a certain value (see W3, W4, and W5), and then begins to increase again (see W6). Furthermore, based on the predicted water retention curves and hydraulic conductivity functions shown in Figure 22, we observe similar results: (a) as the constraint of volumetric soil water content data becomes stronger, the inverted constitutive relationships become closer to the true values provided by the van Genuchten-Mualem model (see W3, W4, and W5); (b) as this constraints continue to increase, the inversion of the constitutive relationship deteriorates (see W6). Additionally, as shown in Figure 21, overfitting occurs when λ_θ is too large. In summary, W5 outperforms other scenarios, making it the default value throughout the paper.

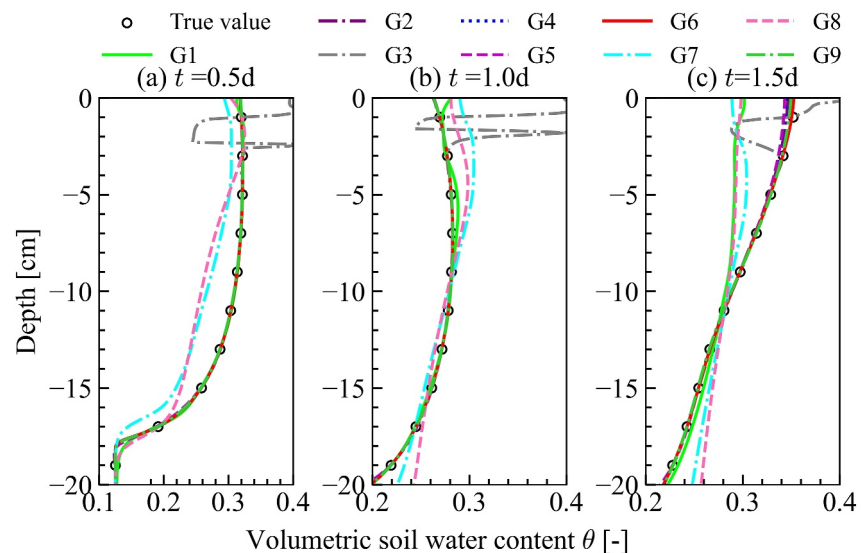


Figure 19. Comparisons of predicted volumetric soil water content by PINN-AWL under the nine different hyper-parameters combinations (G1-G9) and the true values at $t = 0.5, 1, 1.5$ d.

Table 5
The Relative L_2 Error of Volumetric Soil Water Content Predicted by PINN-AWL With Nine Groups of $(\alpha_r, \beta_r, \alpha_\theta, \beta_\theta)$

Scenario	G1	G2	G3	G4	G5	G6	G7	G8	G9
ϵ^θ	0.0777	0.0032	0.1099	0.0050	0.0051	0.0026	0.1350	0.1230	0.0042

6.4. Sensitivity Analysis of Hyperparameter

Through the hyperparameter test conducted above, it becomes apparent that the results are highly responsive to changes in these parameters. Only a handful of appropriate parameters result in the network achieving satisfactory outcomes post-training, while the majority of parameter sets yield unsatisfactory model training. This naturally prompts the consideration of how to select suitable hyperparameters without precise knowledge of the constitutive relationship. Fortunately, it's observed that setting the hyperparameters α_r and α_θ to 10 and 100 allows the network model to effectively simulate water flow across various soils and reverse their constitutive relationships. Loss balance significantly impacts training stability, the efficiency, and the network's capability to accurately represent physical systems. Loss balance is important in physics-informed neural networks (Xiang et al., 2022), a potential reason for unsatisfactory results with inappropriate parameters might be the disruption of balance between loss terms. Whether it's the hyperparameters $(\lambda_\theta, \lambda_r)$ for the balancing loss terms or the hyperparameter $(\alpha_r, \alpha_\theta)$ for adaptively weighted loss, their ratio between the Partial Differential Equation (PDE) loss term and the data loss term is kept at 1:10. Accordingly, in this section, we designate the values for the hyper-parameters λ_θ and λ_r as 10 and 1, respectively, and equate the values of β_r and β_θ to unity., and then the parameter α_r and α_θ , as detailed in Table 8, to observe whether the change of hyper-parameters will lead to the change of prediction results under the loam scenario (S8).

Figures 23 and 24 vividly shows that under the parameter combination in Table 8, the PINN-AWL can well predict the soil volumetric water content and invert the soil constitutive relationship. Parameters groups P1–P3 demonstrate that satisfactory outcomes can be achieved through concurrent decrement or increment of parameters α_r and α_θ . Parameters groups P4 and P5 reveal that when parameter α_θ is held constant and parameter α_r is varied within a specific range ($\pm 50\%$), PINN is capable of yielding satisfactory results. Similarly, Parameters groups P6 and P7 indicate that by fixing parameter α_r and modulating parameter α_θ within an allocated range ($\pm 50\%$), PINN-AWL also can attain satisfactory results. The results presented indicate that acceptable outcomes can be achieved with parameters α_r and α_θ initially set to 10 and 100, respectively, or by making moderate adjustments within a specific range. This suggests that even in the absence of precise working conditions, by assigning values of 10 and 100 to parameters α_r and α_θ , or by varying them within a defined scope ($\pm 50\%$), PINN-

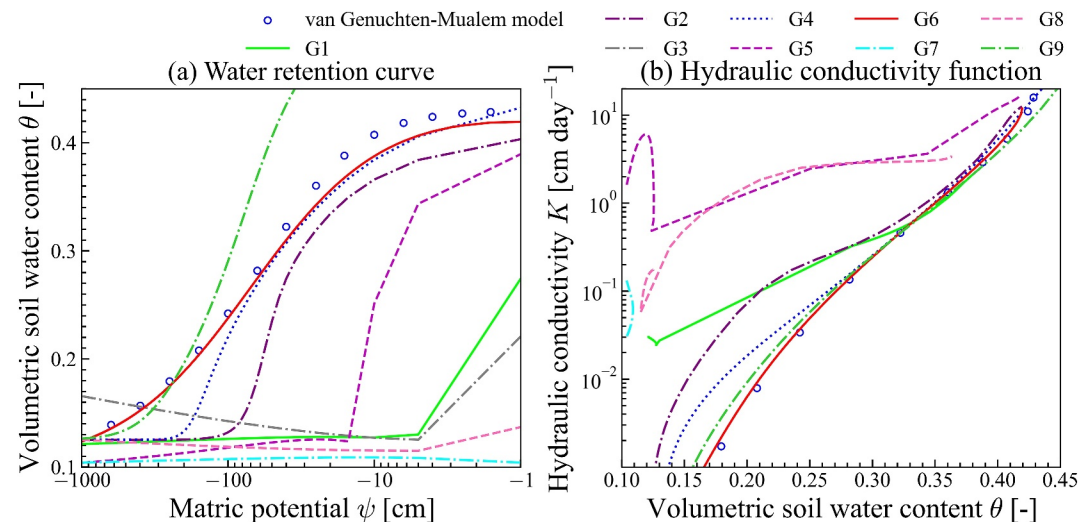


Figure 20. Comparisons of the constitutive relationship: (a) water retention curve and (b) hydraulic conductivity function, from the van Genuchten-Mualem model with those predicted by PINN-AWL under the nine different hyperparameter combinations (G1-G9).

Table 6
Different Groups of Weight Hyper-Parameters in Calculating the Loss Function

Group	λ_θ	λ_r	Group	λ_θ	λ_r	Group	λ_θ	λ_r
W1	1	0	W3	0.1	1	W5	10	1
W2	0	1	W4	1	1	W6	100	1

balance that enhances the model's predictive performance in different scenarios, even when the constitutive relationship is unknown.

7. The PINN-AWL With Sparse or Noisy Training Data

The proposed PINN-AWL takes Richards' equation and a series of volumetric soil water content data to estimate the soil water content, the water retention curve and the unsaturated hydraulic conductivity. The collocation points (t, z) of Richards' equation can be easily generated by classic sampling strategies, but the collection of volumetric soil water content data is usually costly. Moreover, the obtained volumetric soil water content data will inevitably carry the noise. It is necessary to study whether the PINN-AWL still can achieve good prediction even when the volumetric soil water content data is spatially sparse or contains noises.

7.1. Effect of the Spatial Sparsity of Training Data

In practice, volumetric soil water content data is usually collected by sensors arranged at fixed depths, and the cost is usually determined by the number of sensors. That is to say, to reduce the cost of collecting data, we hope to minimize the number of sensors as much as possible without affecting the performance of PINN-AWL. Hence, it is important to study whether PINN-AWL still can obtain accurate predictions when reducing the volumetric soil water content data.

We tested PINN-AWL with loam (scenario S8) when the volumetric soil water content data used from 10 fixed depths ($z = -1, -3, -5, -7, -9, -11, -13, -15, -17, -19$ cm) reduced to seven fixed depths ($z = -1, -4, -7, -10, -13, -16, -19$ cm) and five fixed depths ($z = -1, -5, -9, -13, -17$ cm). The relative L_2 error of the volumetric soil water content at 10 fixed depths, seven fixed depths and five fixed depths were 0.0054, 0.0037, and 0.0046, respectively. The comparisons of the predicted volumetric soil water content by the PINN-AWL using different number of observation depths and the true values at $t = 0.5, 1, 1.5$ d are shown in Figure 25. It can be found that the predicted curves by the PINN-AWL always match well with the true values despite of the reduced amount of data. The comparisons of the predicted constitutive relationships by the PINN-AWL using different number of observation depths with those from the van Genuchten-Mualem model are plotted in Figure 26. The constitutive relationships predicted by the PINN-AWL using the three kinds of fixed depth data show good agreement with the true values, and only small deviations are observed in the matric potential range from $-1,000$ to -100 cm when the five fixed depth data is used. Using the seven fixed depth data shows slight improvement than using the 10 fixed depth data, which is probably due to the fact that the reduced amount of volumetric soil water content data weakens the data constraint, thus allowing Richards' equation to play a greater role.

7.2. Effect of the Noisy Training Data

The data collected by sensors usually contains noises. The capability of the proposed PINN-AWL for dealing with noisy data is investigated by adding Gaussian noises to the simulated volumetric soil water content data generated by Hydrus-1D, that is,

$$\tilde{\theta} = \theta + \sigma \mathcal{N}(0, 1), \quad (17)$$

where $\mathcal{N}(0, 1)$ is the standard Gaussian distribution; θ and $\tilde{\theta}$ are the volumetric soil water content training data with and without noise, respectively, and the noise level is determined by σ . Here, we test for three noise levels, namely, $\sigma = 0.005, 0.01, 0.02$. Loam (scenario S8) is again used as the illustration example. The relative L_2 errors for the predicted volumetric soil water content, matric potential, hydraulic conductivity and soil water fluxes

Table 7
The Relative L_2 Error of Volumetric Soil Water Content Predicted by PINN-AWL With Six Groups of $(\lambda_\theta, \lambda_r)$

Scenario	W1	W2	W3	W4	W5	W6
ϵ^θ	0.0087	0.9724	0.0425	0.0058	0.0039	0.1132

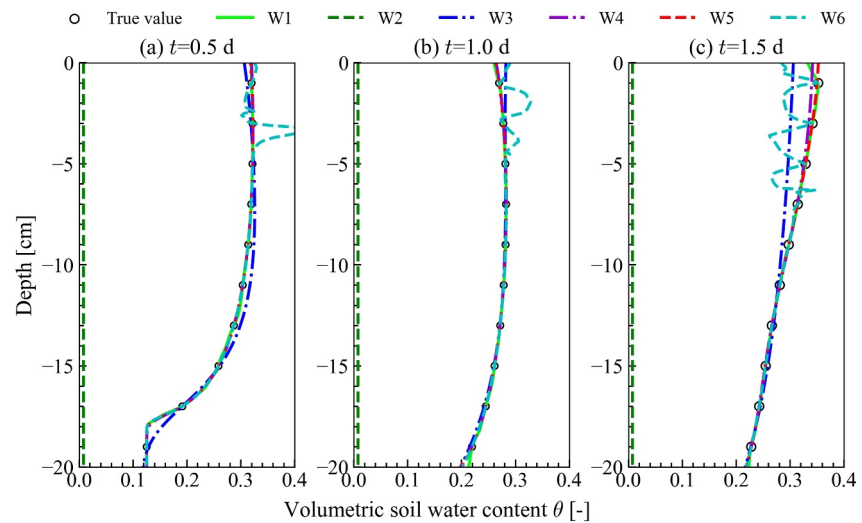


Figure 21. Comparisons of predicted volumetric soil water content by PINN-AWL under the 6 different hyper-parameter combinations (W1-W6) and the true values at different times: (a) $t = 0.5$ d; (b) $t = 1.0$ d; (c) $t = 1.5$ d. The W1 scenario closely aligns with true values but could be overshadowed by other scenarios.

against the ground truth as shown in Table 9 under different noise levels. As the noises gradually increase from 0.005 to 0.02, the relative L_2 error of volumetric soil water content ϵ^θ grows from 0.0056 to 0.013. The predicted volumetric soil water content curves under different levels of noise conditions compared with the true values are shown in Figure 27, and the comparisons of the constitutive relationships from the van Genuchten-Mualem model to those predicted by the PINN-AWL and the distribution of the training data are plotted in Figure 28. The deviations of the predicted relationships can be found as shown in Figure 28, which are much more obvious in the dry end with lower soil water content. Upon careful examination of Figure 28, we found two potential causes for the observed deviations between the predicted and actual curves. One possible reason is the insufficient quantity of volumetric soil water content data in the shifted segments. Another possible reason for the deviations is that minor variations in volumetric water content are associated with more substantial changes in matric potential. This variation relationship, coupled with a small amount of noisy data, may inevitably introduce errors in the matric potential estimation. Therefore, one solution to eliminate bias is to increase the corresponding amount of data, which can be done in practical applications by manually adjusting the data set distribution.

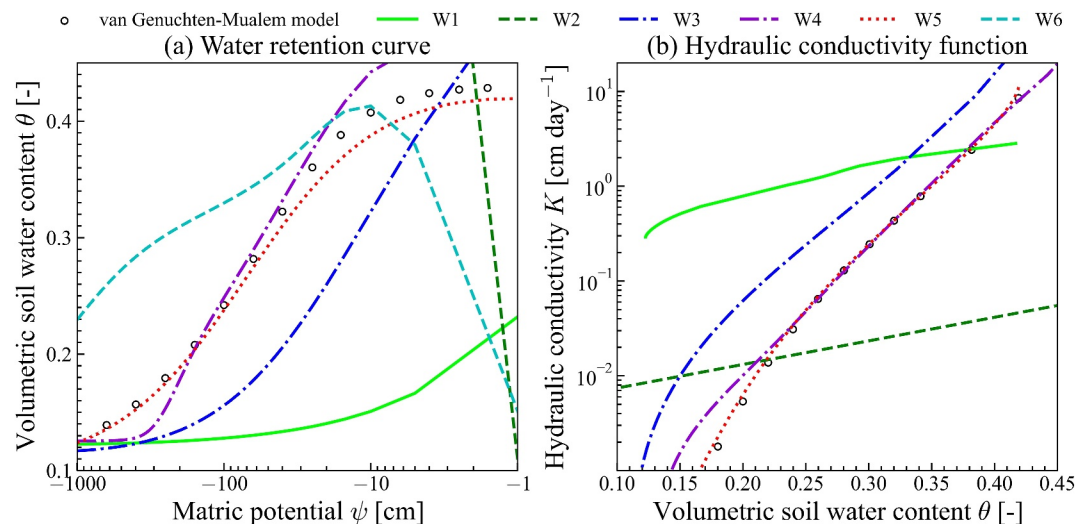


Figure 22. Comparisons of the constitutive relationship: (a) water retention curve and (b) hydraulic conductivity function, from the van Genuchten-Mualem model to those predicted by PINN-AWL under the six different hyper-parameters combinations (W1-W6).

Table 8
Different Combinations of $(\alpha_r, \alpha_\theta)$ for Sensitivity Analysis

Group	α_r	α_θ	Group	α_r	α_θ	Group	α_r	α_θ
P1	5	50	P4	5	100	P7	10	150
P2	7	70	P5	15	100			
P3	15	150	P6	10	50			

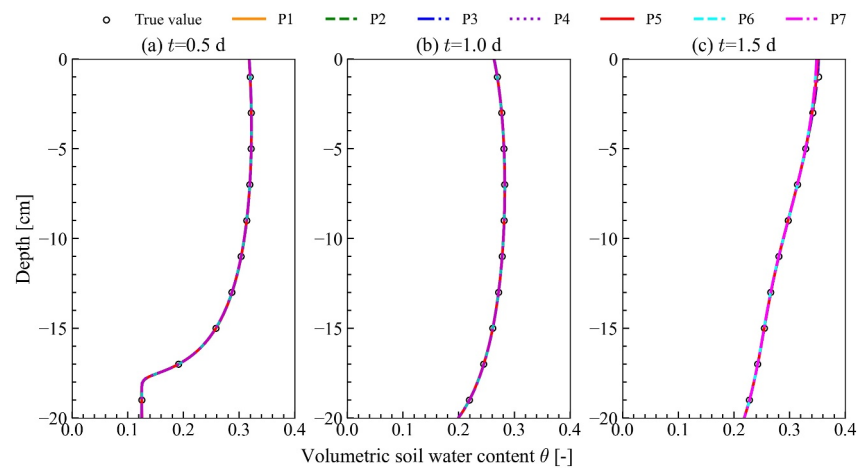


Figure 23. Comparisons of predicted volumetric soil water content by PINN-AWL under the seven different hyper-parameters combinations (P1-P7) and the true values at different times: (a) $t = 0.5$ d; (b) $t = 1.0$ d; (c) $t = 1.5$ d.

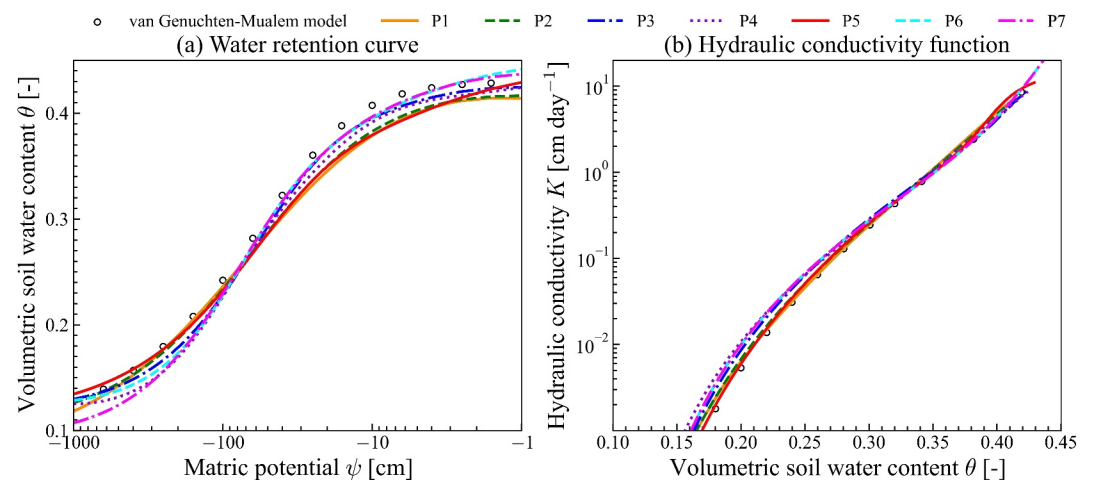


Figure 24. Comparisons of the constitutive relationship: (a) water retention curve and (b) hydraulic conductivity function, from the van Genuchten-Mualem model to those predicted by PINN-AWL under the seven different hyper-parameters combinations (P1-P7).

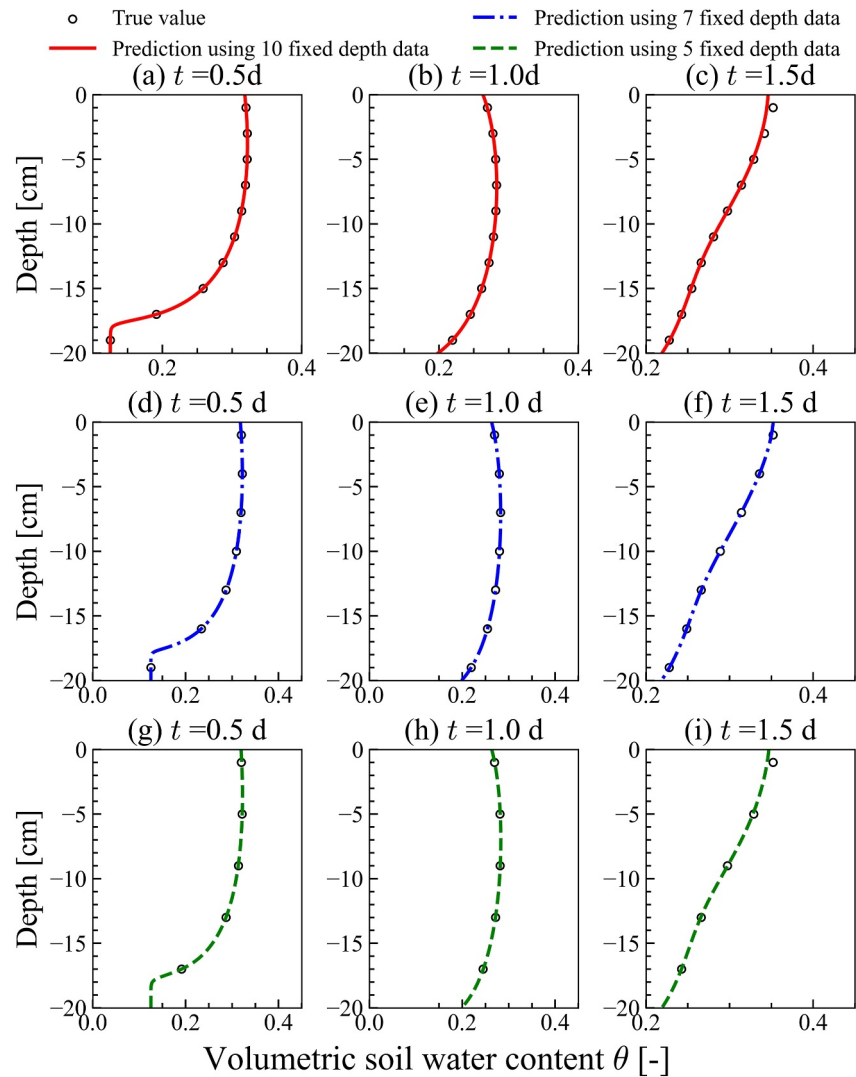


Figure 25. Comparisons of the predicted volumetric soil water content by PINN-AWL using three kinds of different number of observation depths (5, 7, and 10) and the true values for scenario S8 (loam) at $t = -0.5$ d (a, d, g), $t = 1.0$ d (b, e, h), and $t = 1.5$ d (c, f, i).

8. Conclusions

As motivated by the existing work on soil water neural networks, we propose and test in this paper, PINN-AWL, a physics-informed deep learning method with adaptively weighted loss for modeling soil water flows. The results show that our PINN-AWL significantly outperforms both the standard PINN and the monotonic PINN, especially on soils with strong nonlinearity of constitutive relationships. In particular, we have demonstrated through various experiments: (a) the proposed adaptively weighted loss can greatly help improve the model performance by making the PINN-AWL automatically focus more on regions with larger errors, which is very important to produce accurate predictions of the volumetric soil water content and the two constitutive relationships; (b) the PINN-AWL possesses outstanding generalization ability; and (c) the selection of the hyper-parameters ($\alpha_r, \beta_r, \alpha_\theta, \beta_\theta$) for calculating the adaptive weights and hyper-parameters ($\lambda_\theta, \lambda_r$) for calculating the loss function have important effects on the performance of the PINN-AWL. (d) PINN-AW utilizing the hyperparameter set, with ($\alpha_r, \beta_r, \alpha_\theta, \beta_\theta$) equal (10, 1, 100, 1) and ($\lambda_\theta, \lambda_r$) equal (10, 1), not only performs well across various scenarios but also demonstrates robust performance within a range of values, underscoring its versatility. We also would like to note that enhancing the prediction ability under more general and complex conditions, for example, highly heterogeneous soils and multi-physical coupled processes conditions, will be further investigated in our future work.

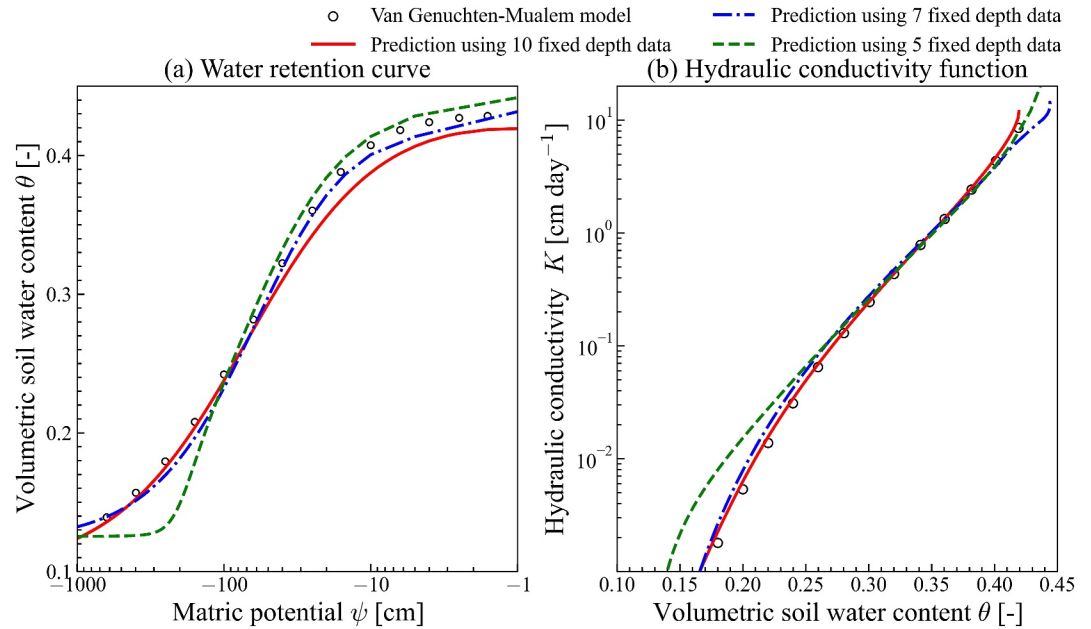


Figure 26. Comparisons of the constitutive relationship: (a) water retention curve and (b) hydraulic conductivity function, from the van Genuchten-Mualem model to the ones predicted by PINN-AWL using the three different kinds of observation depths (5, 7, and 10) for scenario S8 (loam).

Table 9

The Relative L_2 Error of Volumetric Soil Water Content ϵ^θ , Matric Potential ϵ^ψ , Hydraulic Conductivity ϵ^K and Soil Water Flux Density ϵ^q Predicted by PINN-AWL Under Different Noise Level

	ϵ^θ	ϵ^ψ	ϵ^K	ϵ^q
$\sigma = 0.005$	5.6×10^{-3}	8.5×10^{-1}	1.2×10^{-1}	1.1×10^{-1}
$\sigma = 0.01$	7.5×10^{-3}	5.3×10^{-1}	2.2×10^{-1}	1.1×10^{-1}
$\sigma = 0.02$	1.3×10^{-2}	6.3×10^{-1}	3.8×10^{-1}	1.3×10^{-1}

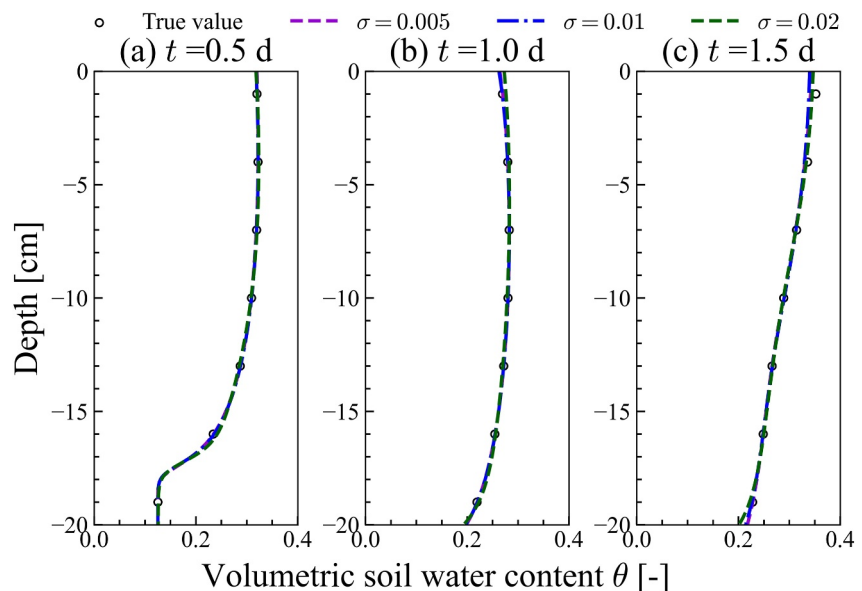


Figure 27. Comparisons of the predicted volumetric soil water content by PINN-AWL under different noise levels ($\sigma = 0.005, 0.01, 0.02$) and the true values for scenario S8 (loam) at different times: (a) $t = 0.5$ d; (b) $t = 1.0$ d; (c) $t = 1.5$ d.

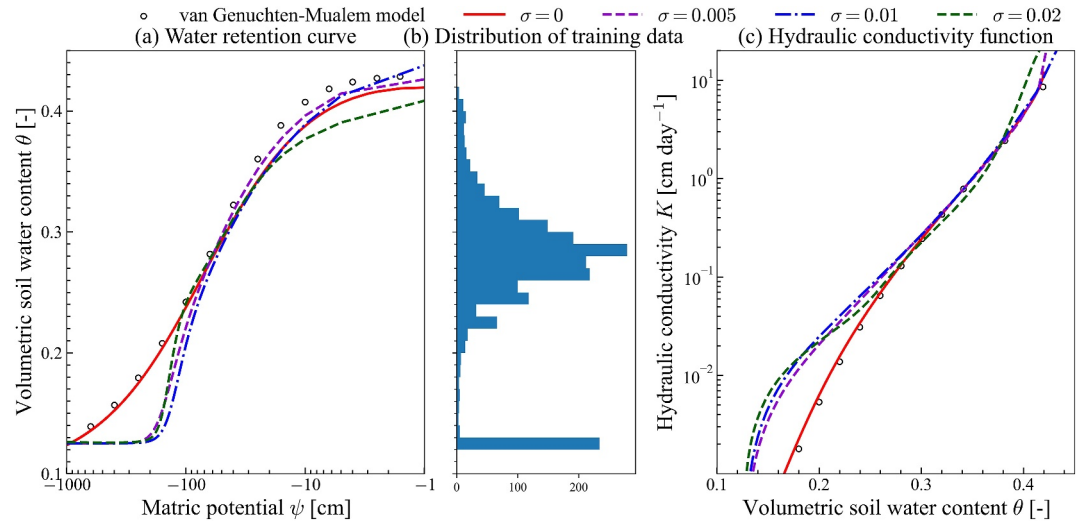


Figure 28. Comparisons of the constitutive relationship (a) water retention curve and (c) hydraulic conductivity function, from the van Genuchten-Mualem model to those predicted by PINN-AWL under different noise levels ($\sigma = 0.005, 0.01, 0.02$) for scenario S8 (loam) and histogram of the training data (b).

Appendix A: Comparisons Between PINN-AWL and Other Two Adaptive Weight Method

In the rapidly evolving field of physics-informed neural networks, various adaptive approaches have been proposed to enhance the learning capabilities and generalization performance. While each method offers unique advantages and insights, it is crucial to systematically evaluate and compare their effectiveness in the context of specific application domains. In this section, we present a comparison of our proposed adaptive network PINN-AWL, with two adaptive networks, which are self-adaptive physics-informed neural networks proposed by McClenny and Braga-Neto (2023) and learning rate annealing physics-informed neural networks proposed by S. Wang et al. (2021).

Firstly, let's briefly introduce two networks. The self-adaptive physics-informed neural networks introduces fully trainable adaptive weights that are applied individually to each training point, enabling the network to identify and focus on the challenging areas of the solution space autonomously. More specifically in this study, the self-adaptation mask function with nonnegative self-adaptation weights will be applied to the volumetric water content data item, so the loss function of SA-PINN in this study is shown as follows:

$$\mathcal{L}_{\text{MSE}}(\mathbf{W}, \mathbf{b}, \boldsymbol{\lambda}) = \frac{1}{2} \sum_{i=1}^{N_\theta} \ell(\hat{\theta}_i) + \frac{1}{2} \sum_{j=1}^{N_r} m(\lambda^j) \ell(\hat{r}_j), \quad (\text{A1})$$

where $\boldsymbol{\lambda} = (\lambda^1, \lambda^2, \dots, \lambda^{N_\theta})$ are trainable, nonnegative self-adaptive weights and self-adaptation mask function $m(\lambda)$ is a nonnegative, differentiable, strictly increasing function of λ . Unlike the parameters of the network model, the trainable parameter λ is updated by gradient ascent, which are:

$$\lambda^{k+1} = \lambda^k + \rho^k \nabla_{\lambda} \mathcal{L}(\mathbf{W}^k, \mathbf{b}^k, \lambda^k), \quad (\text{A2})$$

where:

$$\nabla_{\lambda} \mathcal{L}(\mathbf{W}^k, \mathbf{b}^k, \lambda^k) = \frac{1}{2} \begin{bmatrix} m'(\lambda^{k,1}) |\hat{r}_1|^2 \\ \dots \\ m'(\lambda^{k,N_r}) |\hat{r}_{N_r}|^2 \end{bmatrix}. \quad (\text{A3})$$

By updating through Equation A3, larger loss elements will have greater numerical weights, making the network automatically focus on areas that the solution has larger error. SA-PINN employs both the Adam optimizer and the LBFGS optimizer and it only implements self-adaptation weights method within the Adam optimizer. It is worth noting that the initialization of self-adaptive weights and the selection of self-adaptation mask function affect PINN training. The SA-PINN has approve its effectiveness and accuracy in solving viscous burgers equation, Helmholtz equation, Allen-Cahn reaction-diffusion equation with self-adaptation mask function $m(\lambda) = \lambda^2$. Thus, we used $m(\lambda) = \lambda^2$ in this study.

Learning rate annealing physics-informed neural networks (LA-PINN) is proposed to solve the problem that the numerical stiffness of the unbalanced backpropagation gradient during model training. The key of learning rate annealing physics-informed neural networks is to set the parameter λ_r as 1, while the parameter λ_θ will be recalculated based on the gradient of the different loss terms at each iteration, which is calculated by moving average of the form as follows,

$$\lambda_\theta = (1 - \alpha_1)\lambda_\theta + \alpha_1\hat{\lambda}_\theta, \quad (\text{A4})$$

where α_1 is hyper-parameter, $\hat{\lambda}_\theta$ is the ratio between the maximum gradient value attained by the mean of the gradient magnitudes computed for the loss term as,

$$\hat{\lambda}_\theta = \frac{\max_{(\mathbf{W}, \mathbf{b})} \{ |\nabla_{(\mathbf{W}, \mathbf{b})} \mathcal{L}_r(\mathbf{W}, \mathbf{b})| \}}{|\nabla_{(\mathbf{W}, \mathbf{b})} \mathcal{L}_\theta(\mathbf{W}, \mathbf{b})|} = \frac{\max_{(\mathbf{W}, \mathbf{b})} \left\{ \left| \nabla_{(\mathbf{W}, \mathbf{b})} \frac{1}{N_r} \sum_{j=1}^{N_r} \ell(\hat{r}_j) \right| \right\}}{\left| \nabla_{(\mathbf{W}, \mathbf{b})} \frac{1}{N_\theta} \sum_{i=1}^{N_\theta} \ell(\hat{\theta}_i) \right|}, \quad (\text{A5})$$

After obtaining parameter λ_θ in each round of iterative calculation, all that remains is to calculate new losses and optimization.

The model architectures, training epochs, and other parameters of the three PINNs (PINN-AWL, SA-PINN, LA-PINN) are identical; the only variation among them is the respective loss functions they utilized. The performance of three PINNs in loam example (S8) is shown in Figures A1 and A2. Figure A1 shows the performance of the three models in predicting volumetric water content, and the results show that PINN-AWL and LA-PINN are capable of predicting water content, while SA-PINN can only make a rough estimation. The reason for the low accuracy of SA-PINN may be that the weights are applied to the residual term, thus the network focuses more on the residual term.

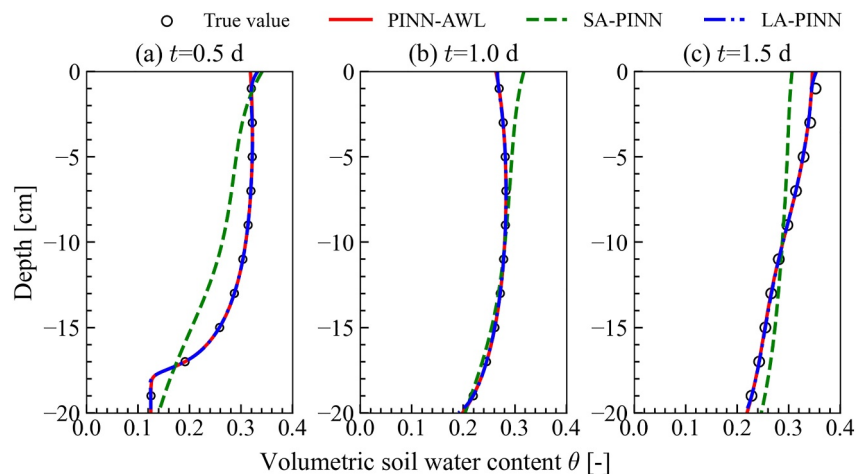


Figure A1. Comparisons of volumetric soil water content predicted by PINN-AWL, SA-PINN (self-adaptive PINN) and LA-PINN (learning rate annealing physics-informed neural networks) and the true values at different times: (a) $t = 0.5$ d; (b) $t = 1.0$ d; (c) $t = 1.5$ d.

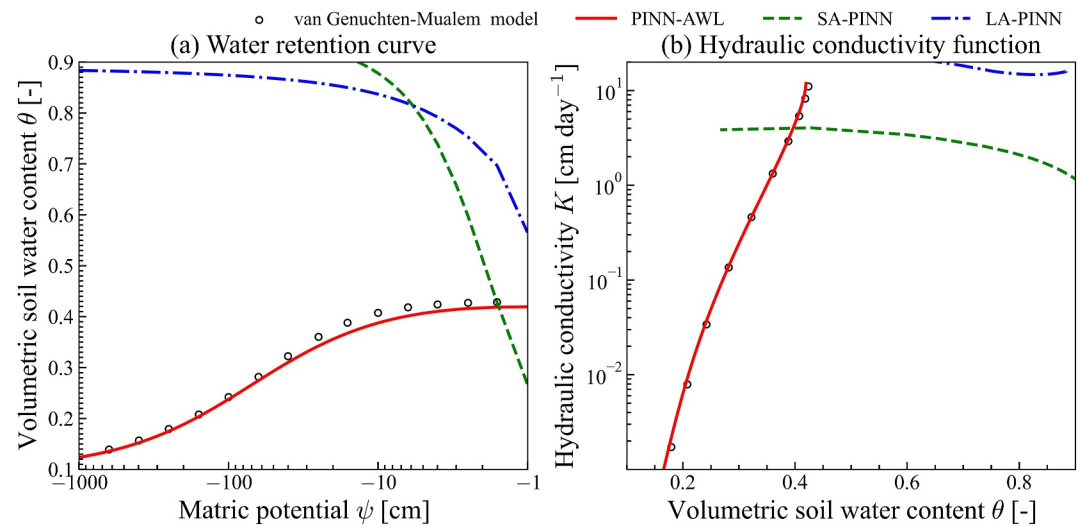


Figure A2. Comparisons of the constitutive relationship: (a) water retention curve and (b) hydraulic conductivity function, predicted by the three PINNs (PINN-AWL, SA-PINN, LA-PINN) and those from the van Genuchten-Mualem model.

As for the constitutive relationship, it becomes apparent that only PINN-AWL successfully inverted the results, while the other two PINNs failed. Several factors could have contributed to this. Firstly, SA-PINN obviously pays more attention to the residual term, and LA-PINN focuses on maintaining the balance between different terms during the gradient descent process, which may weaken the role of the data term, while the appropriate constraints on the data term can help the network to converge to a satisfactory result. Secondly, in the proposed neural network, the initial and boundary conditions are not included, which could result in more challenging for SA-PINN and LA-PINN. In summary, directly applying SA-PINN and LA-PINN to the currently studied network and loss function cannot obtain satisfactory results. However, this does not mean that these two methods are invalid, and it is possible to customize these methods to produce better performance and results.

All the three PINNs change the original loss function and contributes to an increased demand on computational resources. PINN-AWL calculates an adaptive weight for each training sample. SA-PINN not only calculates an adaptive weight for each training configuration point, but also updates these weights using the gradient rise method. Both PINN-AWL and SA-PINN only need to calculate the gradient information of the total loss and update the parameters, while LA-PINN needs to calculate the gradient information of each loss term in the loss function in each iteration to obtain the parameters and then update the network parameters, which increases the calculation burden of each iteration. The above results have proved that PINN-AWL has better robustness, which is also simpler, easier to understand and implement.

Appendix B: The Application of PINN-AWL to Experimental Data

As for application of the PINN-AWL with laboratory experimental or field data, we have applied the PINN-AWL framework to laboratory experimental data from Sadeghi's upward infiltration studies (Sadeghi et al., 2017), using their sand (AZ2) soil moisture data set to demonstrate the method's practical feasibility. The experimental data set originates from constant-pressure upward flow tests on diverse soils, where volumetric soil water content was measured via shortwave infrared (SWIR) camera reflectance. More specific details can be found from Sadeghi et al. (2017) and Bandai et al. (2024). The experimental data is available on Bandai et al. (2023). We adopted their spatial domain definition of $[-0.1, 0]$ m for the infiltration column and utilized measurement points at $(-0.085, -0.075, -0.065, -0.055, -0.045, -0.035, -0.035, -0.015)$ m. The data set was used to train the PINN-AWL model.

As illustrated in Figure B1, PINN-AWL achieves accurate predictions of volumetric soil water content, closely matching the experimental measurements. For constitutive relationship inversion, For the inversion of constitutive relations, we plotted the water retention curve predicted by PINN-AWL alongside experimental

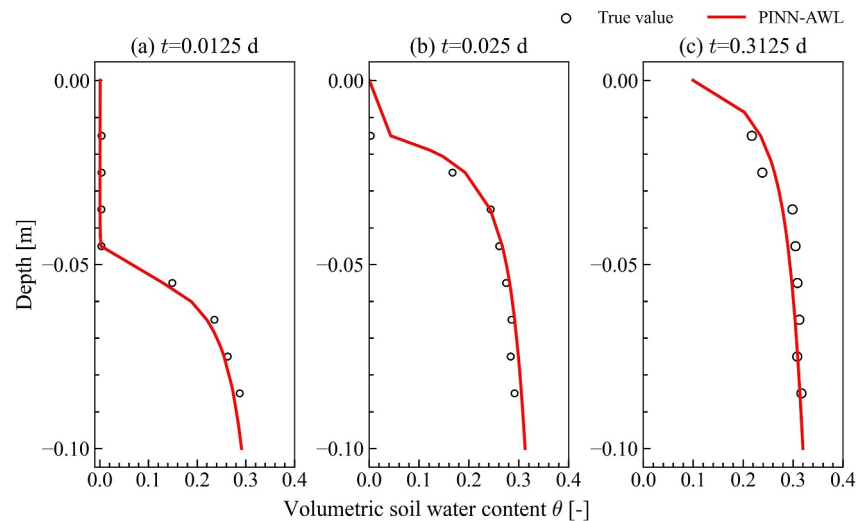


Figure B1. Comparisons of predicted volumetric soil water content θ by PINN-AWL and the true values for experimental data at different times: (a) $t = 0.0125$ d; (b) $t = 0.025$ d; (c) $t = 0.3125$ d.

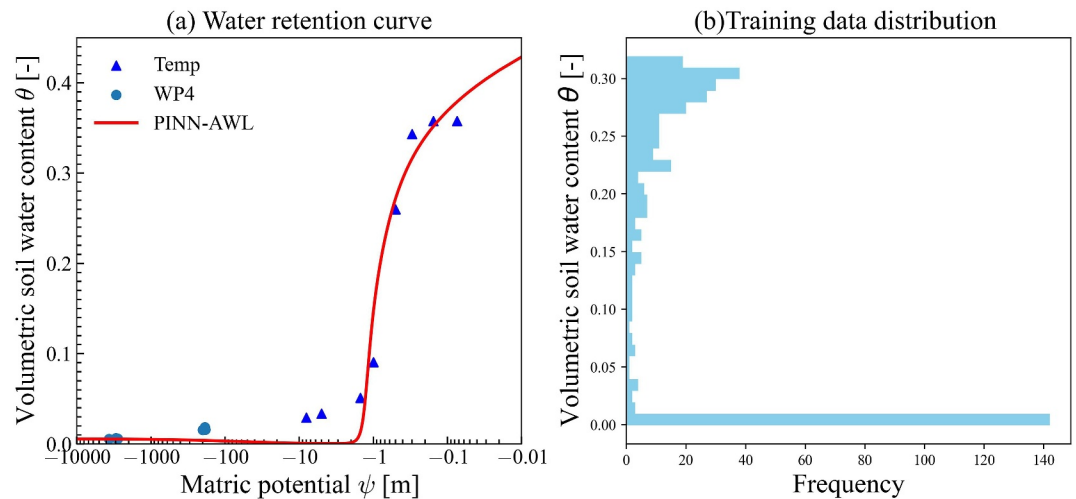


Figure B2. Comparison of water retention curves predicted by PINN-AWL and experimental data, with training data distribution: (a) water retention curve, (b) training data distribution. Blue triangles indicate measurements from Tempe cells (Soilmoisture Equipment Corp., USA), and blue circles from WP4-T Dewpoint Potentiometer (METER Group, Inc., USA).

measurements obtained using Tempe cells (Soilmoisture Equipment Corp., USA) and the WP4-T Dewpoint Potentiometer (METER Group, Inc., USA), as shown in Figure B2. The comparison shows that PINN-AWL provides a reasonable estimation of the water retention curve, though discrepancies are evident at both the dry and wet ends. Possible reasons for these deviations include: (a) at the dry end, volumetric soil water content changes only slightly (from 0 to 0.04), whereas the matric potential varies significantly (from $-10,000$ to -10 m), making accurate prediction difficult with limited data at dry end; (b) at the wet end, the training data lacks corresponding values—the maximum volumetric soil water content in the training set is around 0.31, while the measured value reaches approximately 0.35; (c) noise in the data set may also contribute to the observed errors.

Appendix C: Additional Evaluation on Sand and Clay Soils

To further assess the generalizability of the proposed PINN-AWL framework, we conducted additional tests on two representative soils at the extremes of the texture spectrum: Sand and Clay. The hydraulic properties of these soils were characterized using the van Genuchten-Mualem (VG-M) model. The specific soil hydraulic parameters are provided in Table C1, and the upper boundary condition applied in the simulation is the atmospheric boundary condition, as detailed in Table C2, while the lower boundary is set as a free drainage condition.

The simulations were carried out using Hydrus-1D over a 1-m soil profile and a simulation period of 2 days. Following the simulations, we extracted the volumetric water content data at a fixed depth from the top 20 cm of

Table C1

The Soil Hydraulic Parameters of the van Genuchten-Mualem Model for the Sand and Clay (Carsel & Parrish, 1988)

Soil type	θ_r [–]	θ_s [–]	α [cm ^{–1}]	n [–]	K_s [cm day ^{–1}]	l [–]
Sand	0.045	0.43	0.145	2.68	712.8	0.5
Clay	0.068	0.38	0.008	1.09	4.8	0.5

Table C2

The Upper Boundary Conditions and Soil Types for Two Soils

Soil type	Sand	Clay
Time [day]	Water flux density [cm day ^{–1}]	
0.25	10	1
0.5	10	0.8
0.75	0	0
1	0	0.3
1.25	10	1
1.5	5	0.8
1.75	0	0.3
2	0	0

Note. It should be noted that the positive value represents potential evaporation, and the upward is positive, and downward is negative.

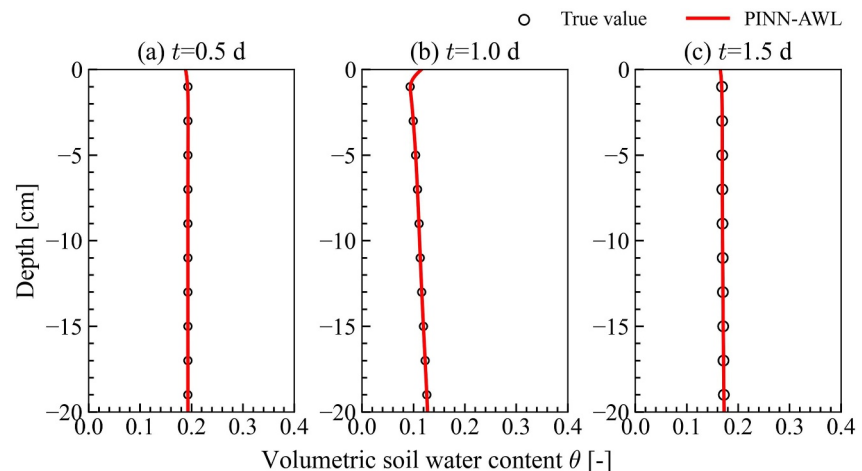


Figure C1. Comparisons of predicted volumetric soil water content by PINN-AWL under sand and the true values at different times: (a) $t = 0.5$ d; (b) $t = 1.0$ d; (c) $t = 1.5$ d.

the soil profile and used it to train the PINN-AWL. The results of the experiments are presented in Figures C1–C4. Figures C1 and C2 illustrate the prediction of volumetric soil water content by PINN-AWL for both sand and clay soils. The model performs well in capturing the dynamics of soil moisture content. Figure C3 shows that PINN-AWL can successfully predict the underlying constitutive relationship for clay soil. However, Figure C4 indicates that PINN-AWL fails to accurately invert the constitutive relationship for sand soil. A possible explanation for the poor inversion performance in sand soil is the high nonlinearity of the water retention curve in the wet range. Despite the large infiltration flux imposed at the surface boundary, the water rapidly percolates through the sand, resulting in minimal changes in the volumetric soil water content within the topsoil. This is evident from the volumetric soil water content of the soil profile at different moments (as shown in Figure C1) and the distribution of the training data as shown in Figure C4b, which lacks sufficient information for effective learning by PINN-AWL.

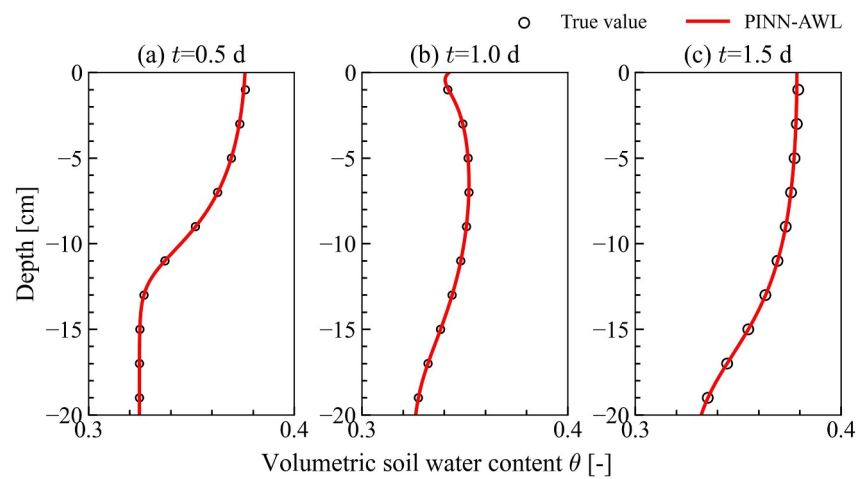


Figure C2. Comparisons of predicted volumetric soil water content by PINN-AWL under clay and the true values at different times: (a) $t = 0.5$ d; (b) $t = 1.0$ d; (c) $t = 1.5$ d.

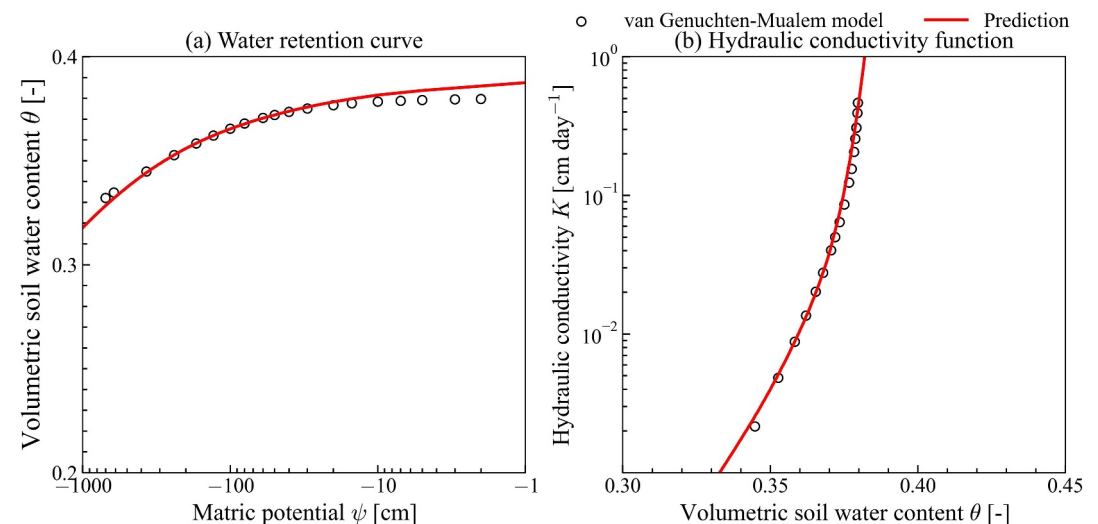


Figure C3. Comparisons of the constitutive relationship: (a) water retention curve and (b) hydraulic conductivity function, predicted by the PINN-AWL and those from the van Genuchten-Mualem model under clay.

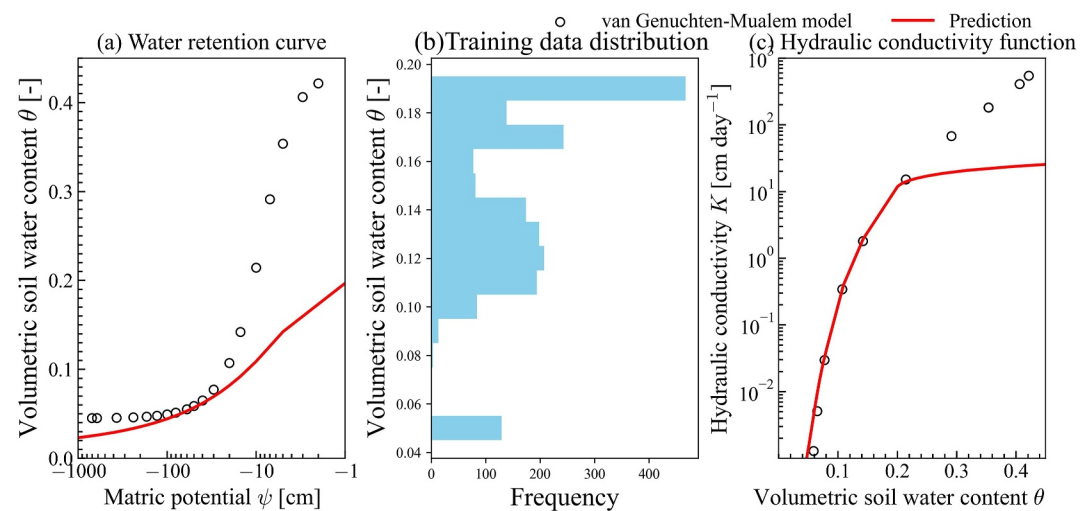


Figure C4. Comparisons of the constitutive relationship: (a) water retention curve and (c) hydraulic conductivity function, predicted by the PINN-AWL and those from the van Genuchten-Mualem model under sand and (b) the distribution of training data.

Data Availability Statement

The data and codes used in this paper are available on Li et al. (2025). The experimental data is available on Bandai et al. (2023).

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